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In partial fulfilment of the requirement for the award
of degree of **Master of Computer Application**
(Discipline -AI and ML)

**TITLE: AI-Based Skill Gap Analysis and Learning
Recommendation System**

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ABSTRACT

The rapid evolution of information technology, artificial intelligence, and data-driven industries has significantly transformed the nature of work and the skills required to succeed in modern professional environments. While educational institutions continue to produce graduates with formal qualifications, employers across the globe increasingly report that many graduates lack the specific technical and analytical skills required to perform effectively in real-world roles. This persistent mismatch between academic learning outcomes and industry expectations has resulted in what is commonly known as the skill gap problem. The skill gap is particularly evident in technology-intensive domains such as software engineering, data analytics, artificial intelligence, and machine learning, where both theoretical understanding and practical competency are essential.

Traditional assessment systems in higher education primarily rely on examinations, grading systems, and cumulative performance indicators. These methods provide a general measure of achievement but often fail to capture the depth of a learner's conceptual understanding, practical skill, and job readiness. Two students may receive similar grades while having very different levels of real-world competence. Furthermore, students themselves are frequently unaware of their weaknesses because feedback is limited to marks or pass/fail status rather than detailed diagnostic insights. As a result, learners often continue studying without addressing their true learning deficiencies, leading to ineffective preparation for professional careers.

With the growth of digital learning platforms, online assessments, and learning management systems, enormous amounts of educational data are being generated. Every quiz attempt, assignment submission, interaction, and time spent on learning materials creates valuable data that can be used to analyze learning behavior. Artificial Intelligence

and Machine Learning provide the tools needed to process this data and extract meaningful insights. These technologies make it possible to identify hidden patterns in learner performance, predict learning outcomes, and personalize education in ways that were previously impossible.

The project titled “AI Based Skill Gap Analysis and Learning Recommendation System” is designed to harness the power of artificial intelligence to address the skill gap problem in technology education. The system aims to provide a data-driven, objective, and personalized method of identifying what a learner knows, what they do not know, and what they need to learn next. It is developed for a broad group of users, including BCA, BTech, MCA, and MTech students, as well as aspiring data analysts, machine learning engineers, and other technology professionals.

The core of the system is an MCQ-based assessment framework. Multiple-choice questions are widely used in education because they allow objective scoring and easy automation. However, in this project, MCQs are used in a more advanced way. Each question is carefully mapped to specific concepts, sub-concepts, and job-related competencies. For example, a question on linear regression may be linked to concepts such as statistics, optimization, and model evaluation, and also mapped to job roles like data analyst or machine learning engineer. By analyzing which questions a learner answers correctly or incorrectly, along with how long they take to respond, the system builds a detailed profile of their knowledge and skills.

Machine learning models are then applied to this data to estimate the learner’s level of mastery for each concept. Instead of simply counting the number of correct answers, the system considers factors such as question difficulty, historical performance, and consistency over time. This results in a more accurate and meaningful representation of skill level. From these mastery scores, the system computes a Skill Gap Index, which

indicates how far a learner is from the expected level of proficiency for a given concept or job role.

One of the most important features of the proposed system is its ability to generate personalized learning recommendations. Once skill gaps are identified, the system suggests specific topics that the learner should focus on. These recommendations are not random; they follow a prerequisite-aware structure, ensuring that learners first strengthen foundational knowledge before moving on to advanced topics. For example, a learner struggling with neural networks may be guided to first revise linear algebra and probability if those are identified as weak areas.

To ensure accessibility and cost-effectiveness, the system emphasizes the use of freely available learning resources. Instead of directing learners only to paid courses, it provides links to open educational materials such as video lectures, tutorials, and online documentation. This makes the system particularly valuable for students who may not have access to expensive training programs.

The system supports two primary learning modes. In concept-based learning mode, users focus on mastering specific topics within a subject, such as programming, data structures, or machine learning algorithms. In role-based learning mode, users select a target job role, such as data analyst or AI engineer, and the system evaluates their readiness for that role by comparing their current skills with industry-aligned competency requirements. This dual-mode approach allows the system to serve both academic learners and job seekers.

From a research perspective, this project combines elements of educational data mining, machine learning, and learning analytics. It not only builds a working system but also evaluates its effectiveness through empirical analysis. By comparing learner performance

before and after using the system, the study assesses whether AI-driven recommendations lead to measurable improvement in skills.

In conclusion, the AI Based Skill Gap Analysis and Learning Recommendation System represents a significant step toward more intelligent, personalized, and industry-relevant education. By transforming raw assessment data into actionable learning guidance, the system helps learners become more self-aware, more focused, and more prepared for their careers. It also provides educators and institutions with valuable insights into student learning patterns, enabling them to improve teaching strategies and curriculum design. Through the integration of artificial intelligence, data analysis, and educational theory, this project demonstrates how technology can be used to bridge the gap between education and employment in the modern digital economy.

DECLARATION

I, Swapnita Paras, a student pursuing Master of Computer Applications (MCA), 4th Semester at Amity University Online, hereby declare that the project work entitled “AI-Based Skill Gap Analysis and Learning Recommendation System” has been prepared by me during the academic year 2024-2026 under the guidance of Mr. Hitesh Gupta, Tech Lead – Artificial Intelligence, Comviva. I assert that this project is a piece of original bona-fide work done by me. It is the outcome of my own effort and that it has not been submitted to any other university for the award of any degree.

A handwritten signature in cursive script that reads "Swapnita".

Signature of Student

CERTIFICATE

This is to certify that Swapnita Paras of Amity University Online has carried out the project work presented in this project report entitled “AI-Based Skill Gap Analysis and Learning Recommendation System” for the award of Master of Computer Applications (MCA) under my guidance. The project report embodies results of original work, and studies are carried out by the student himself/herself. Certified further, that to the best of my knowledge the work reported herein does not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

A handwritten signature in blue ink, appearing to read 'Hitesh', with a stylized flourish extending from the end.

Mr. Hitesh Gupta

Tech Lead – Artificial Intelligence

Comviva

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CHAPTER 1

INTRODUCTION TO THE STUDY

1.1 Introduction

The twenty-first century has witnessed an unprecedented acceleration in technological innovation, driven largely by advances in computing power, data availability, and algorithmic sophistication. Artificial intelligence (AI), machine learning (ML), big data analytics, cloud computing, and software engineering have transitioned from specialized research domains into mainstream drivers of economic growth and organizational competitiveness. These technologies are now deeply embedded across sectors including healthcare, finance, manufacturing, education, governance, logistics, and retail. Consequently, the nature of work has evolved significantly, demanding new forms of expertise and adaptability from the modern workforce.

In this evolving professional landscape, higher education institutions are expected to prepare graduates who possess not only theoretical knowledge but also the ability to apply concepts effectively in real-world contexts. Employers increasingly emphasize practical skills, analytical reasoning, problem-solving ability, and familiarity with contemporary tools and workflows. However, despite ongoing curricular reforms and pedagogical innovations, a substantial mismatch persists between academic learning outcomes and industry expectations. This mismatch is commonly referred to as the skill gap.

The skill gap represents the difference between the competencies learners acquire through formal education and the competencies required to

perform effectively in professional roles. In technology-intensive fields, this gap is especially critical because competencies are hierarchical and interdependent. Advanced capabilities in areas such as deep learning, predictive modeling, and intelligent system design depend heavily on strong foundations in mathematics, statistics, programming, and algorithmic thinking. When these foundational competencies are weak or fragmented, learners struggle to progress, even when exposed to advanced material. Traditional assessment systems in higher education rely predominantly on written examinations, assignments, and cumulative grades as measures of achievement. While these methods provide standardized and scalable evaluation mechanisms, they often lack diagnostic precision. Grades typically aggregate performance across multiple topics, masking specific areas of weakness. Consequently, two learners with similar grades may possess vastly different mastery profiles. Moreover, assessment feedback is often limited to numerical scores, offering little insight into how learners can improve.

The limitations of conventional assessment systems are further exacerbated by the scale and diversity of modern education. Large class sizes, heterogeneous learner backgrounds, and constrained instructional resources make individualized feedback difficult to deliver. As a result, many learners progress through academic programs without a clear understanding of their conceptual gaps, leading to inefficient study strategies and reduced employability.

At the same time, the digital transformation of education has generated vast amounts of learner interaction data. Online assessments, learning

management systems, and interactive platforms record detailed information about learner behavior, including response accuracy, time-on-task, error patterns, and learning sequences. Artificial intelligence and machine learning provide powerful tools to analyze this data, enabling fine-grained diagnostics, predictive insights, and personalized learning interventions.

The present project, titled “AI Based Skill Gap Analysis and Learning Recommendation System,” seeks to harness these technological capabilities to design an intelligent framework for diagnosing skill gaps and guiding learners toward effective remediation. By focusing on concept-level mastery and prerequisite-aware learning paths, the system aims to enhance learner self-awareness, optimize study efforts, and improve readiness for industry-aligned roles.

1.2 Nature of Skill Gaps in AI and Technical Education

The rapid evolution of artificial intelligence and data-driven technologies has fundamentally altered the nature of professional competence in technical fields. Unlike traditional disciplines where knowledge structures remain relatively stable over long periods, AI and machine learning are characterized by continuous innovation, frequent paradigm shifts, and rapidly expanding tool ecosystems. New frameworks, programming libraries, algorithms, and best practices emerge at a pace that far exceeds the rate at which formal academic curricula can be updated. As a result, even students who perform well in university examinations may graduate with knowledge that is already partially outdated or misaligned with current industry requirements. This creates a form of skill gap that is not merely a result of poor learning, but of structural lag between education systems and technological change.

Skill gaps in AI education are also multidimensional. They do not simply reflect a lack of knowledge, but often arise from an imbalance between conceptual understanding, practical implementation, and analytical reasoning. Many learners acquire surface-level familiarity with algorithms or tools through memorization or guided tutorials, yet lack the deeper comprehension needed to adapt those tools to new problems. For example, a student may be able to train a machine learning model using a predefined library, but may not fully understand why certain hyperparameters matter, how data distributions influence model behavior, or how to diagnose overfitting. Such gaps remain hidden in conventional testing environments but become critically important in real-world professional settings.

Another defining feature of AI-related skill gaps is the hierarchical nature of knowledge. Competence in advanced topics such as deep learning, reinforcement learning, or natural language processing depends heavily on mastery of foundational subjects including linear algebra, probability theory, statistics, optimization, and programming. If learners possess weak or fragmented understanding in these prerequisite areas, their learning trajectory becomes unstable. They may experience repeated failure or frustration when confronted with advanced material, even if they previously achieved acceptable grades. This phenomenon illustrates that skill gaps are not isolated deficiencies but interconnected weaknesses within a broader cognitive structure.

Furthermore, the diversity of learner backgrounds exacerbates the skill gap problem. Students entering AI and data science programs often come from varied educational pathways, ranging from mathematics and engineering to business, humanities, or self-taught programming. While this diversity enriches the learning environment, it also creates uneven baseline competencies. Some learners may excel in coding but

struggle with statistical reasoning, while others may understand theory but lack implementation skills. Without systematic diagnostic mechanisms, these differences remain unaddressed, causing some students to fall behind while others move ahead without adequate challenge.

The traditional assumption that progression through a syllabus equates to mastery is particularly problematic in technical education. Passing a course in machine learning does not necessarily mean that a learner can design, evaluate, and improve a model in a real-world context. Assessment methods that emphasize final answers rather than the reasoning process further obscure true competence. As a result, learners may accumulate academic credits without developing the robust skill profiles required by employers, leading to frustration among graduates and dissatisfaction among industry stakeholders.

In addition to cognitive and curricular factors, motivational and behavioral aspects also contribute to skill gaps. Many learners struggle with time management, ineffective study strategies, or lack of confidence when facing complex quantitative material. When such challenges are not detected early, they can compound over time, resulting in disengagement and attrition. AI-based diagnostic systems have the potential to identify not only what learners know, but how they learn, enabling more targeted and supportive interventions.

Understanding the nature of skill gaps in AI and technical education therefore requires a holistic perspective that integrates cognitive, curricular, behavioral, and technological dimensions. These gaps are not simply failures of individual learners; they are systemic mismatches between how knowledge is taught, assessed, and applied. Recognizing this complexity is essential for designing intelligent systems

capable of accurately diagnosing learner needs and guiding them toward effective, personalized learning pathways.

1.3 Weaknesses of Current University Assessment Systems

Assessment systems play a central role in shaping how students learn, what they prioritize, and how their competence is judged. In most universities, particularly in technology and engineering disciplines, assessment is still dominated by written examinations, end-semester tests, periodic assignments, and cumulative grading schemes. While these methods offer administrative convenience and allow institutions to evaluate large cohorts of students efficiently, they suffer from significant conceptual and pedagogical limitations. These limitations are especially damaging in fields such as artificial intelligence, data science, and machine learning, where deep understanding, problem-solving ability, and adaptability are more important than rote recall.

One of the primary weaknesses of conventional assessment is its heavy reliance on summative evaluation. Students are typically evaluated at fixed points in time, such as midterm and final examinations, which determine most of their grades. This approach treats assessment as a judgment of performance rather than as a tool for learning. Consequently, students often focus on maximizing scores rather than on developing long-term understanding. In technical subjects, this leads to strategic memorization of formulas, code templates, and standard problem types, while neglecting the conceptual reasoning that underpins them.

Another major problem is that most assessments provide only coarse-grained feedback. A final score or grade may indicate that a student achieved, for example, 70% or a “B” in a course, but it does not reveal which specific concepts were mastered and which were misunderstood. In an AI course, two students with the

same grade may have very different skill profiles: one may be strong in probability but weak in programming, while another may be proficient in coding but lack statistical intuition. Traditional grading systems collapse these distinctions into a single number, thereby concealing important diagnostic information that could guide learning and remediation.

The design of assessment items also contributes to the persistence of skill gaps. Many university exams rely on predictable problem patterns that reward procedural execution rather than conceptual insight. Students who learn how to recognize these patterns can perform well without truly understanding the underlying principles. In machine learning, for instance, a student may learn to apply a particular algorithm to a familiar dataset type without understanding why that algorithm is appropriate or how it would behave under different conditions. This kind of superficial competence often fails when learners encounter novel or complex real-world problems.

Scalability further complicates the issue. With growing student populations and limited instructional resources, instructors often have little time to provide individualized feedback. Grading large numbers of assignments and exams makes it impractical to analyze each student's conceptual weaknesses in detail. As a result, feedback tends to be generic, delayed, and focused on errors rather than on learning strategies. Students may see where they lost marks, but not how to systematically improve their understanding.

Moreover, conventional assessments are poorly equipped to capture dynamic learning processes. Learning in AI and technical subjects is not a linear accumulation of facts but a process of iterative refinement, hypothesis testing, and conceptual restructuring. Students often experience sudden breakthroughs or persistent misconceptions that are invisible in static test scores. Without continuous and fine-

grained measurement, educators cannot detect when a learner is stuck, when they are guessing, or when they have developed incorrect mental models.

The growing availability of digital learning environments highlights the inadequacy of traditional assessment even more clearly. Online platforms can capture rich data about how learners interact with content, how long they spend on questions, how often they revise answers, and what kinds of mistakes they make. However, most universities still rely on legacy assessment frameworks that ignore this data and focus only on final outcomes. This represents a missed opportunity to transform assessment into a powerful diagnostic and learning support tool.

These weaknesses collectively contribute to the persistence of skill gaps in higher education. Students may advance through programs, accumulate credits, and even graduate with respectable grades, yet remain unprepared for the demands of AI-driven professional environments. Addressing this problem requires a shift from coarse, summative evaluation toward continuous, data-driven, and concept-level assessment—an approach that lies at the heart of the AI-based system proposed in this study.

1.4 Role of Data, Learning Analytics, and Student Behavior

The increasing digitization of education has transformed the learning process into a data-rich activity. Learning management systems, online quizzes, intelligent tutoring platforms, and virtual laboratories generate vast volumes of information about how learners interact with educational content. Every click, response, submission, revision, and pause produces data that can be analyzed to reveal patterns of engagement, understanding, and struggle. In the context of artificial intelligence and technical education, this data represents an invaluable resource for uncovering the underlying causes of skill gaps and designing more effective learning interventions.

Learning analytics refers to the systematic collection, measurement, and analysis of learner data for the purpose of understanding and optimizing learning. Unlike traditional assessments, which capture only final outcomes, learning analytics focuses on the process of learning itself. For example, it can reveal how long a student spends on a particular concept, how often they revisit a topic, whether they guess or reason through questions, and how their performance evolves over time. These behavioral signals provide insights that are far more granular and informative than a single exam score.

In AI and machine learning education, learning analytics is particularly important because conceptual difficulties often manifest in subtle ways. A student may answer a question correctly but take an unusually long time, indicating uncertainty or fragile understanding. Another student may answer quickly but make systematic errors that reveal a misunderstanding of a key principle. By analyzing response times, error patterns, and learning sequences, an intelligent system can distinguish between confident mastery, superficial familiarity, and genuine confusion. This allows educators and systems to move beyond binary notions of right and wrong toward a richer understanding of learner competence.

Behavioral data also sheds light on motivation and self-regulation, which are critical factors in technical learning. Students who frequently abandon questions, skip challenging material, or avoid practice exercises are at higher risk of developing skill gaps, even if they initially perform well. Learning analytics can detect such patterns early, enabling timely interventions such as targeted feedback, encouragement, or adaptive difficulty adjustment. This proactive approach contrasts sharply with traditional systems, which often discover problems only after a student has already failed an exam.

The integration of learning analytics into AI-driven systems also supports personalization. Different learners progress at different speeds and require different types of support. Some may benefit from additional practice, while others need conceptual explanations or alternative examples. By continuously analyzing learner data, an intelligent system can adapt its recommendations to the evolving needs of each individual. This dynamic personalization is especially valuable in complex subjects like machine learning, where misunderstandings can easily cascade into larger conceptual failures.

Another important aspect of learning analytics is its ability to reveal population-level trends. By aggregating data across large numbers of learners, institutions can identify which topics are most difficult, which instructional materials are most effective, and where curricula may need improvement. For instance, if a large proportion of students consistently struggle with gradient descent or probability distributions, this may indicate a need for better instructional design or prerequisite reinforcement. Thus, learning analytics not only supports individual learners but also informs systemic educational reform.

Despite its potential, learning analytics is still underutilized in many educational settings. Data is often collected but not effectively analyzed or acted upon. This gap between data availability and data-driven decision-making highlights the need for intelligent systems that can automatically process large datasets and translate insights into actionable recommendations. The AI-based skill gap analysis framework proposed in this study is designed to fulfill precisely this role by converting raw learner data into meaningful diagnostic and instructional guidance.

1.5 Machine Learning for Skill Diagnosis

Machine learning provides a powerful methodological foundation for transforming raw educational data into meaningful insights about learner competence. In traditional educational systems, diagnosing a student's strengths and weaknesses relies heavily on human judgment, limited test scores, and subjective interpretation. While experienced instructors can often identify struggling students, such judgments are inherently constrained by time, class size, and the availability of observable evidence. Machine learning overcomes these limitations by enabling automated, data-driven analysis of learner behavior at a scale and level of precision that would be impossible through manual evaluation.

At the core of skill diagnosis lies the problem of inferring latent knowledge states from observable actions. When a learner answers a multiple-choice question, submits a piece of code, or interacts with an online tutorial, they reveal partial information about their understanding. However, these observable behaviors do not directly correspond to underlying competence. A correct answer may result from genuine understanding, educated guessing, or prior exposure to a similar problem, while an incorrect answer may arise from a simple mistake or a deeper misconception.

Machine learning models are well suited to disentangle these possibilities by analyzing large numbers of interactions across many learners and contexts.

In AI and technical education, this diagnostic challenge is especially complex because knowledge is structured hierarchically and probabilistically. A student's ability to solve a regression problem depends not only on their knowledge of regression itself, but also on their understanding of linear algebra, statistics, and data preprocessing. Machine learning models such as Bayesian knowledge tracing, item response theory extensions, and neural network-based student models can capture these dependencies by representing learner knowledge as a set of

probabilistic states that evolve over time. As students practice and receive feedback, these models update their estimates of mastery, providing a continuously refined picture of each learner's skill profile.

One of the key advantages of machine learning-based diagnosis is its ability to incorporate multiple sources of evidence. Instead of relying solely on whether an answer is correct, models can also consider response time, the number of attempts, patterns of errors, and even the sequence in which topics are studied. For example, a learner who consistently takes longer to answer questions on probability theory may be flagged as having low confidence or incomplete understanding, even if their accuracy is relatively high. This multidimensional perspective allows for more sensitive and accurate detection of learning gaps.

Another important benefit is adaptability. Machine learning models can be trained and refined using new data, enabling them to improve over time as more learners interact with the system. This is particularly valuable in rapidly evolving fields such as AI, where new topics and tools continually emerge. By learning from ongoing usage, diagnostic models can remain relevant and accurate, ensuring that recommendations reflect current learning patterns and content structures.

Machine learning also enables early intervention. By analyzing trends in a learner's performance, models can predict future difficulties before they become severe. For instance, if a student shows declining accuracy or increasing response times on prerequisite topics, the system can infer a growing risk of failure in upcoming advanced material. Such predictive insights allow for timely remedial action, reducing the likelihood of long-term skill gaps and disengagement.

Through these mechanisms, machine learning transforms assessment from a static snapshot into a dynamic, evolving representation of learner competence. Instead of

waiting for end-of-course exams to reveal problems, educators and learners gain continuous, fine-grained feedback about progress and challenges. This shift is essential for supporting mastery in complex, fast-moving technical domains and forms a central pillar of the AI-based skill gap analysis system developed in this study.

1.6 Knowledge Graphs and Prerequisite Modeling

A fundamental challenge in technical education is that knowledge is not a flat collection of independent topics, but a structured network of interrelated concepts. In artificial intelligence and machine learning, this structure is especially complex.

Understanding a single advanced topic, such as neural networks or support vector machines, depends on prior mastery of multiple foundational ideas, including linear algebra, probability theory, calculus, optimization, and data representation. If these prerequisites are weak or missing, learning becomes inefficient and error-prone.

Knowledge graphs and prerequisite modeling provide a systematic way to represent and manage these dependencies.

A knowledge graph is a formal representation of concepts and the relationships between them. In an educational context, nodes represent individual concepts or skills, while edges represent prerequisite or dependency relationships. For example, a node representing “logistic regression” may be connected to nodes for “probability,” “linear algebra,” and “gradient descent.” By organizing content in this way, educators and intelligent systems can explicitly model the pathways through which learners should progress to achieve mastery. This makes it possible to move beyond linear curricula and toward more flexible, learner-centered learning structures.

Prerequisite modeling is the process of identifying and encoding which concepts must be learned before others can be effectively understood. In traditional curricula,

these relationships are often implicit or loosely defined. Students may be told that they need to study mathematics before machine learning, but the specific mathematical skills required are rarely mapped in detail. Knowledge graphs make these relationships explicit, allowing both learners and systems to see exactly which foundational concepts support which advanced topics. This clarity is crucial for diagnosing skill gaps and planning effective remediation.

In an AI-based learning system, knowledge graphs serve as the backbone for adaptive guidance. When a learner performs poorly on a particular topic, the system can trace backward through the graph to identify which prerequisite concepts may be responsible. For instance, if a student struggles with clustering algorithms, the system may determine that the root cause lies in weak understanding of distance metrics or probability distributions. Rather than providing generic advice, the system can then recommend targeted review of the specific foundational concepts that need strengthening.

Knowledge graphs also support efficient learning by preventing unnecessary or premature exposure to advanced material. Without prerequisite modeling, learners may attempt to study complex topics before they are ready, leading to confusion and frustration. By using the graph structure to assess readiness, the system can guide learners along pathways that are appropriate for their current level of mastery. This ensures that time and effort are invested where they will have the greatest educational impact.

Another advantage of knowledge graphs is their ability to evolve. As new topics, tools, and methodologies emerge in the AI field, they can be added to the graph along with their prerequisite relationships. This allows the educational framework to remain up to date without requiring a complete redesign of the curriculum. In this

way, knowledge graphs provide a flexible and scalable foundation for lifelong learning in rapidly changing technical domains.

By integrating knowledge graphs with machine learning-based diagnostic models, the proposed system in this study creates a powerful synergy. Diagnostic models estimate a learner's mastery of individual concepts, while the knowledge graph uses this information to determine the most appropriate next steps. Together, they enable precise, personalized, and pedagogically sound learning recommendations that address the root causes of skill gaps rather than merely their symptoms.

1.7 Recommendation Systems in Education

Recommendation systems are widely known for their use in e-commerce, streaming platforms, and social media, where they suggest products, movies, or content based on user preferences and behavior. In education, however, recommendation systems serve a much more profound and transformative purpose. Rather than optimizing for engagement or consumption, educational recommendation systems aim to optimize for learning, understanding, and long-term skill development. In the context of artificial intelligence and technical education, such systems play a crucial role in guiding learners through complex and often overwhelming bodies of knowledge.

Traditional educational pathways are typically fixed and uniform, assuming that all learners progress at roughly the same pace and in the same sequence. This assumption is rarely true, especially in AI-related fields where students come from diverse backgrounds and possess varying levels of prior knowledge. Some learners may need extensive reinforcement in mathematics, while others may require more practice in programming or data interpretation. Recommendation systems allow learning pathways to become adaptive rather than static, tailoring content to the specific needs of each individual.

An educational recommendation system works by combining information about a learner's current knowledge state with a model of the learning domain. Using data from assessments, interactions, and performance history, the system estimates what the learner knows, what they are struggling with, and what they are ready to learn next. It then selects and suggests learning resources—such as videos, readings, exercises, or projects—that are most likely to help the learner make progress. In an AI-based skill gap analysis system, these recommendations are not arbitrary but are grounded in diagnostic evidence and prerequisite relationships.

One of the major benefits of recommendation systems is efficiency. Learners often waste significant time studying material that is either too easy, too difficult, or irrelevant to their actual weaknesses. By directing learners to precisely the content they need, recommendation systems help them focus their effort where it will have the greatest impact. This is particularly important in technical subjects, where cognitive overload and frustration are common when learners are presented with poorly matched material.

Another critical feature of effective educational recommendation systems is transparency and explainability. Unlike commercial platforms that often make opaque recommendations, learning systems must earn the trust of users. When a system suggests that a learner should review probability theory or practice matrix operations, it should also explain why these topics are important and how they relate to the learner's goals. This not only increases acceptance of the recommendations but also helps learners develop metacognitive awareness of their own learning process.

Recommendation systems also support continuous improvement by adapting to changes in learner behavior. As learners study, practice, and improve, their mastery

levels change, and so should the system's guidance. A dynamic recommendation engine can update its suggestions in real time, ensuring that learners are always working on material that is aligned with their current needs. This ongoing adaptation helps maintain motivation and prevents stagnation.

In the proposed AI-based system, recommendation algorithms are integrated with machine learning diagnostics and knowledge graphs to provide coherent, personalized learning pathways. Rather than simply listing resources, the system constructs structured sequences of learning activities that build from foundational concepts to advanced applications. This transforms the learning experience from a fragmented exploration of content into a guided journey toward mastery and professional readiness.

1.8 Equity, Access, and Scalable Learning

One of the most significant challenges in modern education is ensuring that opportunities for high-quality learning are accessible to all learners, regardless of their socio-economic background, geographic location, or institutional affiliation. In highly technical fields such as artificial intelligence and data science, this challenge is particularly acute. Advanced training programs, specialized instructors, and high-quality learning materials are often concentrated in elite institutions or urban centers, leaving many capable learners without adequate support. As a result, skill gaps are not only academic but also social and economic in nature.

AI-based learning systems offer a powerful mechanism for addressing these inequalities by making expert-level guidance scalable and widely available. Unlike human instructors, who are limited by time and class size, intelligent systems can provide continuous, personalized feedback to thousands or even millions of learners simultaneously. This scalability enables students from smaller colleges, rural areas,

or self-learning backgrounds to receive the same quality of diagnostic insight and learning recommendations as those in well-resourced institutions. In this way, technology can act as a leveling force, reducing disparities in educational opportunity.

Access to learning resources is another critical dimension of equity. Many traditional education models depend on expensive textbooks, proprietary software, or paid training programs, which can exclude learners who cannot afford them. An AI-based recommendation system that prioritizes freely available or open educational resources can significantly lower these barriers. By directing learners to high-quality, no-cost materials such as open textbooks, online lectures, and public datasets, the system ensures that financial constraints do not prevent motivated individuals from developing valuable technical skills.

Language, learning style, and prior educational experience also influence how effectively learners engage with technical material. Some students may benefit from visual explanations, while others prefer step-by-step problem-solving or hands-on projects. An intelligent learning system can accommodate this diversity by offering multiple types of resources and adapting recommendations based on observed learner behavior. This flexibility supports inclusive education by respecting the varied ways in which individuals learn and process information.

Scalability also plays an important role in maintaining educational quality as demand for AI-related skills continues to grow. The global workforce increasingly requires competence in data analysis, automation, and intelligent systems, yet traditional educational infrastructures struggle to keep pace. By automating key functions such as assessment, diagnosis, and guidance, AI-based systems can support large populations of learners without sacrificing personalization. This is essential for

meeting the rising demand for technical expertise in a sustainable and equitable manner.

Finally, equitable learning is not only about providing access but also about ensuring that learners receive appropriate support to succeed. Without diagnostic tools, disadvantaged learners may struggle silently, falling further behind without knowing why. AI-based skill gap analysis systems can identify these learners early and provide targeted assistance, helping to prevent small gaps from becoming insurmountable barriers. In this way, technology can promote not only efficiency but also fairness and social mobility.

1.9 Research Gap and Justification for the Study

Despite significant advances in educational technology, learning analytics, and artificial intelligence, a substantial gap remains between theoretical research and practical implementation in real educational environments. Many existing systems focus on isolated aspects of learning, such as automated grading, adaptive quizzes, or content recommendation, but few provide an integrated framework that combines diagnostic accuracy, pedagogical structure, and actionable learning guidance. This fragmentation limits the effectiveness of current solutions and leaves important educational needs unmet, particularly in complex and rapidly evolving fields such as artificial intelligence and machine learning.

One major research gap lies in the limited ability of existing assessment systems to diagnose concept-level mastery. Traditional examinations and many online testing platforms provide only coarse indicators of performance, such as total scores or percentages. While some adaptive testing systems attempt to adjust difficulty, they often lack a deep understanding of the conceptual relationships among topics. As a result, they can identify that a learner is struggling, but not precisely why. Without

this insight, recommendations tend to be generic and may fail to address the root causes of learning difficulties.

Another gap is the insufficient integration of prerequisite knowledge modeling into learning systems. Although educational research has long emphasized the importance of building on foundational concepts, many digital learning platforms still treat topics as relatively independent units. This leads to learning paths that may be logically inconsistent or pedagogically inefficient. Learners are often directed to advanced material without having secured the necessary groundwork, resulting in confusion, frustration, and disengagement. There is a clear need for systems that explicitly model and utilize prerequisite relationships to guide learners in a structured and meaningful way.

Existing recommendation systems in education also tend to prioritize content delivery over learning effectiveness. Many platforms suggest resources based on popularity, user ratings, or superficial similarity to past activity, rather than on a rigorous analysis of a learner's knowledge state. While such approaches may increase engagement, they do not guarantee that learners are working on the concepts that will most improve their competence. This highlights the need for recommendation engines that are driven by diagnostic evidence and grounded in educational theory.

Furthermore, while learning analytics has made it possible to collect large volumes of data about student behavior, this data is often underutilized. Many institutions lack the tools or expertise needed to transform raw interaction logs into meaningful insights and interventions. As a result, the potential of data-driven education remains largely unrealized. An integrated AI-based system that automatically analyzes learner

data and translates it into personalized guidance would represent a significant step forward.

The present study addresses these gaps by proposing and developing an AI-based Skill Gap Analysis and Learning Recommendation System that unifies machine learning diagnostics, knowledge graph-based prerequisite modeling, and adaptive recommendation algorithms within a single coherent framework. By focusing on concept-level mastery, explainable recommendations, and equitable access to learning resources, the system aims to provide both learners and educators with a powerful tool for improving educational outcomes in AI and related technical fields.

This integrated approach is not only innovative but also urgently needed in a world where the demand for advanced technical skills continues to grow. By bridging the divide between assessment and instruction, and by aligning educational processes more closely with industry-relevant competencies, the proposed system seeks to make a meaningful contribution to both educational research and practical workforce development.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

A rigorous review of literature forms the intellectual backbone of any scholarly investigation by providing theoretical grounding, contextual relevance, and methodological justification. In the domain of technology education and workforce readiness, existing literature consistently identifies a growing disconnect between what students learn in academic programs and the competencies required by industry. This gap has become increasingly visible with the rapid advancement of artificial intelligence, machine learning, data analytics, and software-driven systems, all of which continuously redefine professional skill requirements.

Educational institutions today are evaluated not only by the knowledge they transmit, but also by their ability to produce graduates who are adaptable, competent, and ready to contribute immediately in professional environments. However, most traditional pedagogical models continue to prioritize syllabus completion, time-based progression, and summative examinations. As a result, many graduates complete their degrees without acquiring the depth of conceptual understanding or applied competence demanded by modern technological roles.

These limitations have motivated scholars to explore alternative approaches such as competency-based education, diagnostic assessment, and AI-driven learning analytics.

This chapter synthesizes literature from several interrelated domains that directly inform the present study. These include theories of skills and competencies, competency-based education frameworks, skill gap theory, learning theories underlying diagnostic assessment, and critiques of conventional evaluation systems.

Together, these perspectives establish a strong academic foundation for the development of an AI-based skill gap analysis and learning recommendation system.

2.2 Conceptualization of Skills in Educational Literature

The concept of “skill” has undergone significant transformation within educational research. Early interpretations viewed skills as isolated, task-specific abilities acquired through repetition and training. Such definitions were well suited to industrial-era employment, where work roles were stable and procedures were routine. In contrast, modern knowledge economies require a more holistic understanding of skills as complex, multidimensional capabilities.

Contemporary literature defines skills as an integration of declarative knowledge, procedural ability, cognitive strategies, and contextual judgment. In technology education, these dimensions interact continuously. For instance, programming proficiency involves not only syntactic knowledge of a language, but also algorithmic reasoning, debugging skills, and the ability to apply solutions in real-world contexts. Research consistently shows that knowledge of syntax alone, without conceptual understanding, produces fragile and easily transferable competence.

Skills are also widely recognized as hierarchical and interdependent. Foundational abilities support intermediate competencies, which in turn enable advanced performance. In artificial intelligence and machine learning, high-level tasks such as model optimization or deployment depend critically on mathematics, probability, statistics, and data preprocessing. Weaknesses in these foundations propagate upward, undermining advanced learning. This hierarchical view of skills directly informs assessment design and curriculum structure, making it essential to evaluate and remediate foundational concepts before learners progress further.

2.3 Competency-Based Education Frameworks

Competency-Based Education (CBE) has emerged as a major response to the limitations of traditional time-based academic models. CBE emphasizes clearly defined learning outcomes, mastery-based progression, and alignment with workplace performance standards. Instead of assuming that learning occurs through exposure to content over fixed time periods, CBE requires learners to demonstrate competence before advancing.

Research on CBE highlights several benefits, including improved transparency, greater flexibility for learners from diverse backgrounds, and stronger alignment with industry expectations. By tying progression to demonstrated mastery, competency-based approaches also enhance learner motivation and accountability.

However, scholars also identify major challenges in implementing CBE at scale. Developing reliable, valid, and fine-grained assessment systems requires substantial resources and technological infrastructure. Many institutions struggle to provide continuous diagnostics and personalized feedback, resulting in partial or symbolic adoption of CBE principles. In technical fields, competency frameworks often remain too coarse to support personalized learning. This study addresses that limitation by operationalizing competencies at the concept level through AI-based diagnostics.

2.4 Skill Gap Theory: Academic and Industry Perspectives

The concept of the skill gap is widely discussed in academic research, public policy, and employer surveys. In academic terms, a skill gap reflects misalignment between educational outcomes and occupational demands. From an industry

perspective, it appears as graduates who hold formal qualifications but lack practical, job-ready capabilities.

Literature distinguishes several forms of skill gaps. Vertical gaps involve insufficient depth of knowledge, where learners possess only superficial understanding.

Horizontal gaps involve imbalance across domains, such as strong theory but weak technical execution. Temporal gaps arise when skills become obsolete due to rapid technological change. In AI-driven fields, all three forms are especially pronounced.

The consequences of skill gaps extend beyond individual learners to organizations and economies. Employers incur high retraining costs, and productivity suffers.

These findings reinforce the importance of early, concept-level diagnosis and continuous feedback during learning—precisely the role of the AI-based system proposed in this research.

2.5 Diagnostic Assessment in Educational Research

The concept of diagnostic assessment occupies a central position in modern educational theory. Unlike summative assessment, which evaluates learning after instruction has been completed, diagnostic assessment is designed to identify learners' strengths, weaknesses, and misconceptions before and during the learning process. Educational researchers emphasize that diagnostic assessment enables instructors and systems to tailor instruction to individual learner needs rather than relying on uniform teaching approaches. In technology education, where conceptual dependencies are strong and misconceptions can propagate across topics, diagnostic assessment is particularly critical.

Studies in science, mathematics, and engineering education demonstrate that early detection of misconceptions significantly improves learning outcomes. When learners

receive timely feedback about specific conceptual gaps, they are more likely to adjust their study strategies and avoid reinforcing incorrect mental models. In contrast, delayed or generalized feedback allows misunderstandings to persist, making later remediation more difficult. This body of literature supports the idea that assessment should function as a learning tool rather than merely a grading mechanism.

In AI and machine learning education, diagnostic assessment is especially important because many concepts are abstract and counterintuitive. For example, learners may struggle with probabilistic reasoning, gradient-based optimization, or high-dimensional data representations. Without targeted diagnostic tools, these difficulties often remain hidden until learners fail complex tasks or examinations. The literature therefore advocates for fine-grained, concept-level diagnostics that reveal not only whether a learner is incorrect, but why.

2.6 Learning Analytics and Educational Data Mining

Learning analytics and educational data mining have emerged as powerful fields focused on extracting actionable insights from learner data. These disciplines analyze patterns in assessment responses, time-on-task, interaction sequences, and engagement metrics to understand how learning unfolds. Research shows that such data-driven approaches can predict student success, identify at-risk learners, and uncover ineffective instructional designs.

In technology-enabled learning environments, every learner interaction generates data that can be leveraged for diagnostic purposes. Studies indicate that combining accuracy data with behavioral indicators such as response time and revision frequency yields more reliable estimates of learner understanding than accuracy

alone. For example, fast incorrect responses often indicate guessing, while slow correct responses may reflect fragile understanding. These insights allow learning systems to distinguish between mastery and superficial performance.

Educational data mining further enables the discovery of latent patterns that are not visible to human instructors. By analyzing large datasets, machine learning algorithms can detect which concepts are most difficult, which sequences of learning lead to better outcomes, and how misconceptions evolve over time. This literature provides strong support for integrating analytics into adaptive learning systems.

2.7 Student Modeling and Learner Profiling

Student modeling refers to the process of representing a learner's knowledge, skills, preferences, and behaviors within an intelligent system. Educational researchers have long argued that effective personalization requires accurate and dynamic learner models. Rather than treating all students as identical, adaptive systems maintain individualized profiles that evolve as learners interact with content.

In technical domains, student models often represent mastery at the concept level rather than at the course level. This allows systems to capture fine-grained differences between learners who may have similar overall scores but very different strengths and weaknesses. For example, two students may both achieve 70% on a machine learning test, but one may be strong in probability while the other excels in programming. Literature shows that such distinctions are crucial for meaningful adaptation.

Modern student modeling approaches integrate cognitive, behavioral, and temporal information. By combining what learners know with how they learn and how their performance changes over time, these models provide a comprehensive picture of

learner development. This literature directly informs the design of AI-based skill gap analysis systems.

2.8 Adaptive Learning Systems

Adaptive learning systems aim to modify instructional content, difficulty, and sequencing based on learner performance. Research in this area demonstrates that adaptation leads to higher engagement, faster mastery, and reduced dropout rates, particularly in complex subjects. Unlike static learning environments, adaptive systems continuously adjust to learner needs.

In AI education, adaptive systems are especially valuable because learners progress at different rates and encounter different obstacles. Some students struggle with mathematical foundations, while others need more practice applying algorithms. Adaptive systems can respond to these differences by selecting appropriate learning activities and pacing.

The literature emphasizes that effective adaptation depends on accurate diagnosis and domain modeling. Without reliable information about what a learner knows and what concepts depend on one another, adaptation becomes arbitrary. This reinforces the importance of integrating diagnostic models and knowledge structures within adaptive learning platforms.

2.9 Feedback, Metacognition, and Self-Regulated Learning

Research consistently highlights the importance of feedback in promoting deep learning. Effective feedback is specific, timely, and aligned with learning goals. In contrast, vague or delayed feedback has little impact on learner improvement. In

technical subjects, feedback that explains why an answer is wrong is particularly important for correcting misconceptions.

Metacognition—the ability to reflect on one’s own learning—is also strongly linked to academic success. Learners who understand what they know and what they do not know are better able to plan, monitor, and adjust their study strategies.

Diagnostic and explainable AI systems support metacognition by making learner progress visible and interpretable.

Self-regulated learning, which involves goal setting, monitoring, and reflection, is especially important in self-paced and online environments. Literature shows that intelligent systems that provide diagnostic feedback and learning recommendations can significantly enhance self-regulation, leading to improved outcomes.

2.10 Explainable and Trustworthy Educational AI

As AI becomes more deeply embedded in education, issues of transparency and trust become increasingly important. Researchers argue that learners and educators are more likely to adopt AI-based systems when they understand how decisions are made. Explainable AI allows users to see why a system diagnosed a particular skill gap or recommended a specific learning resource.

Trustworthy systems are also essential for ethical and pedagogical reasons. When learners believe that recommendations are fair, accurate, and aligned with their goals, they are more likely to engage with the system. Literature warns that opaque algorithms may undermine trust and exacerbate inequalities.

Therefore, modern educational AI must balance predictive accuracy with interpretability. This principle directly guides the design of the AI-based skill gap analysis and learning recommendation system proposed in this study.

2.11 Knowledge Tracing Models in Educational Assessment

Knowledge tracing has emerged as a significant area of research within educational data mining, focusing on modeling a learner's knowledge state over time. The primary objective of knowledge tracing models is to estimate the probability that a learner has mastered a particular concept based on their historical performance. One of the earliest and most influential approaches is Bayesian Knowledge Tracing (BKT), which represents knowledge as a latent binary variable that evolves through learning and forgetting processes.

The literature highlights the strengths of BKT in terms of interpretability and theoretical grounding. However, researchers also note limitations, including assumptions of skill independence and fixed learning parameters. These assumptions may not hold in complex domains such as artificial intelligence and machine learning, where concepts are highly interdependent. As a result, several extensions and alternatives to traditional BKT have been proposed.

Deep Knowledge Tracing (DKT) represents a more recent approach that leverages recurrent neural networks to model learner knowledge states. While DKT has demonstrated strong predictive performance, critics argue that its lack of interpretability limits its usefulness in educational contexts. The literature increasingly emphasizes the need for models that balance predictive accuracy with transparency, particularly when diagnostic decisions affect learner outcomes.

2.12 Cognitive Diagnostic Models

Cognitive Diagnostic Models (CDMs) provide a framework for diagnosing multiple

underlying skills required to answer assessment items correctly. Models such as DINA and DINO represent items as requiring specific combinations of attributes, enabling fine-grained diagnosis of learner mastery profiles. Literature suggests that CDMs are particularly effective for identifying detailed skill gaps in structured domains.

Despite their advantages, CDMs present practical challenges. They require extensive item tagging and large datasets to achieve reliable estimates. Additionally, their complexity can hinder scalability in real-world educational settings. Researchers therefore advocate for hybrid approaches that combine diagnostic precision with computational efficiency.

The present study draws from CDM principles by incorporating concept-level tagging and mastery estimation without relying on overly complex attribute structures.

2.13 Item Response Theory and Assessment Calibration

Item Response Theory (IRT) has long been used to model the relationship between learner ability and item difficulty. Unlike classical test theory, IRT provides item-level parameters that support adaptive testing and assessment calibration. Literature demonstrates that IRT-based models improve measurement precision and fairness.

In diagnostic contexts, IRT is often combined with concept tagging to enhance interpretability. However, pure IRT models typically focus on overall ability rather than concept-specific mastery. As such, researchers have explored hybrid models that integrate IRT with diagnostic frameworks.

2.14 Artificial Intelligence–Based Recommendation Systems in Education

Recommendation systems have been extensively studied in domains such as e-commerce and entertainment, and their application to education has gained increasing attention. Educational recommender systems aim to guide learners toward appropriate resources, activities, or learning paths based on their needs.

The literature categorizes recommendation approaches into collaborative filtering, content-based filtering, and knowledge-based systems. While collaborative filtering leverages similarities among learners, it may ignore pedagogical dependencies.

Knowledge-based systems that incorporate prerequisite relationships are more aligned with educational principles.

Empirical studies indicate that prerequisite-aware recommendation systems yield superior learning outcomes compared to generic recommendations. Learners benefit from structured guidance that reinforces foundational concepts before introducing advanced material.

2.15 Ethical Considerations and Responsible Use of AI in Education

The application of artificial intelligence in education raises important ethical concerns, including data privacy, algorithmic bias, and transparency. Literature emphasizes the need for responsible data governance, informed consent, and fairness-aware modeling.

Researchers caution that opaque algorithms may undermine trust and exacerbate inequalities. Consequently, explainability and human oversight are increasingly viewed as essential components of AI-driven educational systems.

2.16 Summary of Chapter 2

This second part of the literature review examined knowledge tracing models, cognitive diagnostic approaches, item response theory, AI-based recommendation systems, and ethical considerations. Together with Part 1, this chapter provides a comprehensive theoretical and empirical foundation for the methodology and system design presented in the subsequent chapter.

CHAPTER 3

RESEARCH OBJECTIVES AND METHODOLOGY

3.1 Introduction

This chapter constitutes the methodological backbone of the present research and provides an exhaustive, research-intensive, and academically rigorous explanation of how the AI-Based Skill Gap Analysis and Learning Recommendation System has been conceptualized, engineered, and empirically evaluated. The purpose of this chapter is not merely to describe procedural steps, but to justify, contextualize, and critically ground every methodological decision within accepted research practices in computer science, educational data mining, and learning analytics. The chapter has been deliberately written at a level of depth and clarity expected of a postgraduate dissertation in the Master of Computer Applications (MCA) program, with explicit attention to reproducibility, transparency, and evaluative robustness.

The rapid acceleration of Artificial Intelligence (AI) and Machine Learning (ML) technologies has resulted in an unprecedented demand for technically skilled professionals. Despite the proliferation of academic programs and online courses, a persistent disconnect exists between formal instruction and demonstrable job readiness. Learners frequently acquire fragmented knowledge, lacking a coherent understanding of prerequisite dependencies among concepts. Traditional assessment mechanisms—primarily summative in nature—typically compress performance into aggregate scores that obscure specific conceptual weaknesses. Such approaches provide limited diagnostic value and fail to guide learners toward targeted remediation.

This research addresses the aforementioned gap by proposing a system that reconceptualizes assessment as an intelligent diagnostic process. Using MCQ-based

assessments enriched with behavioral metadata, the system estimates fine-grained mastery probabilities at the concept level and translates these estimates into structured, prerequisite-aware learning recommendations. The methodological framework integrates principles from psychometrics, machine learning, knowledge representation, and instructional design. Each methodological layer—from data collection to model validation—has been carefully designed to support both analytical rigor and pedagogical relevance.

This chapter systematically documents the research aim, objectives, design framework, operationalized research problem, scope and delimitations, variable definitions, data sources, sampling strategy, instrument development, data collection protocols, data preprocessing methods, modelling strategies, recommendation engine architecture, experimental evaluation plan, reliability and validity considerations, ethical safeguards, implementation planning, and risk mitigation strategies. Collectively, these sections establish a coherent and defensible methodological foundation for the results presented in subsequent chapters.

3.2 Restatement of Research Aim and Objectives

3.2.1 Research Aim

The central aim of this research is to design, implement, and rigorously evaluate an artificial intelligence–driven system capable of diagnosing fine-grained skill gaps at both conceptual and occupational role levels using MCQ-based assessment data, and to generate personalized, prerequisite-aware learning recommendations using freely available educational resources.

This aim reflects a shift from conventional assessment paradigms toward adaptive, data-informed learning systems. Rather than treating assessment as an endpoint for

grading, the proposed system treats assessment as a continuous feedback mechanism that informs personalized learning interventions. The system is designed to operate at scale, accommodate heterogeneous learner profiles, and produce interpretable outputs that can be meaningfully acted upon by learners and educators alike.

3.2.2 Research Objectives

To operationalize the stated research aim, the study is guided by the following expanded and interrelated objectives:

1. To conceptualize and construct a hierarchical knowledge taxonomy for AI/ML and adjacent computing domains, capturing conceptual granularity, prerequisite dependencies, and conceptual progression from foundational to advanced levels, and to align this taxonomy with role-specific competency expectations for common entry-level technical positions.
2. To design, author, and validate a large-scale MCQ item bank in which each item is systematically annotated with metadata including primary concept, secondary concepts, difficulty level, cognitive level (mapped to Bloom's taxonomy), and explicit prerequisite relationships, thereby enabling fine-grained diagnostic inference.
3. To collect, structure, and preprocess learner response data obtained through controlled assessments, incorporating both real participant data and carefully governed synthetic augmentation to mitigate sparsity and imbalance without compromising realism or analytical integrity.
4. To develop interpretable machine learning and probabilistic models that estimate per-concept mastery probabilities for individual learners, aggregate these estimates into a standardized Skill Gap Index, and infer role-readiness indicators aligned with predefined competency benchmarks.

5. To engineer a knowledge-graph-driven recommendation system that utilizes prerequisite structures to generate sequenced, personalized remediation pathways and dynamically selects pedagogically appropriate, freely accessible learning resources.
6. To empirically evaluate the diagnostic accuracy, educational effectiveness, and usability of the system through controlled experimentation, statistical analysis of learning gains, and structured qualitative feedback from learners.

3.3 Research Design

The research employs a mixed-methods design that integrates system-oriented engineering research with empirical educational evaluation. This design is particularly well-suited to interdisciplinary studies at the intersection of computer science and education, where both artifact construction and impact assessment are essential.

3.3.1 Phase A: Design and Implementation (Engineering Phase)

Phase A focuses on the iterative design and implementation of the complete system architecture. This includes the development of the assessment delivery platform, the construction of the concept taxonomy and role mappings, the creation of the MCQ item bank, the design of the data logging and storage infrastructure, the implementation of feature engineering and modelling pipelines, and the development of the recommendation engine.

During this phase, the research follows a design science methodology, wherein artifacts are built, evaluated, and refined through successive iterations. Emphasis is placed on modular system design to ensure that individual components can be independently tested and modified without cascading failures. All modelling

assumptions, feature definitions, and architectural decisions are explicitly documented to ensure methodological traceability.

3.3.2 Phase B: Empirical Evaluation (Quantitative and Qualitative Phase)

Phase B involves the empirical evaluation of the developed system under controlled conditions. This phase assesses not only predictive performance but also educational impact and user experience. Participants are exposed to the system through structured pre-tests, intervention periods, and post-tests. Quantitative data is collected to measure changes in mastery probabilities, while qualitative data is gathered through surveys and brief interviews to capture perceptions of usefulness, clarity, and trust.

The integration of quantitative and qualitative evidence strengthens the validity of the research findings and provides a holistic assessment of system effectiveness.

3.4 Operationalization of the Research Problem

The research problem is operationalized through the following guiding question:

Given MCQ response logs enriched with behavioral metadata such as response time, attempt history, and topic sequence information, and a validated mapping between assessment items and a hierarchical concept taxonomy, can machine learning-based models generate reliable per-concept mastery estimates that (a) accurately identify skill gaps relevant to predefined job roles, and (b) when used to drive prerequisite-aware learning recommendations, lead to statistically significant and educationally meaningful improvements in learner mastery?

This operational formulation directly informs the selection of variables, modelling approaches, and evaluation metrics used throughout the study, ensuring alignment between research questions and analytical methods.

3.5 Scope and Delimitations

3.5.1 Scope of the Study

The scope of this research is confined to the assessment and remediation of technical and cognitive skills within AI/ML and closely related computing domains. The system evaluates knowledge across programming fundamentals, data structures, linear algebra, probability and statistics, data preprocessing techniques, supervised and unsupervised learning algorithms, model evaluation methodologies, SQL fundamentals, and introductory deployment concepts.

Assessment is conducted exclusively through MCQs designed to test not only factual recall but also application, interpretation, and analytical reasoning. The focus on MCQs is motivated by their scalability, ease of automated analysis, and suitability for large-scale data collection.

3.5.2 Delimitations of the Study

The study intentionally excludes open-ended programming tasks, project-based assessments, live coding platforms, collaborative teamwork evaluations, and affective or soft-skill dimensions. Furthermore, the role taxonomy is limited to a predefined set of entry-level technical roles and does not attempt to generalize findings to senior or highly specialized positions.

3.6 Variables and Measures

A clear and precise definition of variables is essential for ensuring analytical rigor, interpretability of results, and alignment between research questions and empirical methods. In the present study, variables are categorized into dependent variables,

independent (predictor) variables, and control variables. Each category has been carefully selected to capture different dimensions of learner behavior, system performance, and educational impact.

3.6.1 Dependent Variables

The dependent variables represent the primary outcomes of interest produced by the proposed AI-based system. These variables quantify learner knowledge, identify skill deficiencies, and measure the effectiveness of the learning interventions.

Concept Mastery Probability

Concept mastery probability is defined as a continuous value ranging between 0 and 1, representing the estimated likelihood that a learner has adequately mastered a specific concept. This estimate is generated by the predictive models using a combination of correctness data, behavioral features, and item metadata. Unlike traditional percentage scores, mastery probability reflects uncertainty and confidence in the estimate, enabling more nuanced diagnostic interpretation.

Skill Gap Index

The Skill Gap Index is a derived metric scaled between 0 and 100, designed to quantify the magnitude of discrepancy between a learner's current mastery level and the proficiency level required for a given concept or job role. The index aggregates mastery probabilities across relevant concepts and weights them according to role importance. A higher Skill Gap Index indicates greater remediation need.

Post-Intervention Mastery Gain

Post-intervention mastery gain is calculated as the difference between pre-test and post-test mastery probabilities. This variable serves as the primary indicator of educational impact and is used to evaluate the effectiveness of the recommendation engine.

3.6.2 Independent (Predictor) Variables

Independent variables are those features used by the models to infer mastery probabilities. These variables capture both performance outcomes and behavioral signals.

Item Correctness

Item correctness is a binary variable indicating whether a learner selected the correct answer for a given MCQ. While correctness remains a foundational indicator of knowledge, it is interpreted in conjunction with other contextual features rather than in isolation.

Response Time

Response time, measured in seconds, captures the duration between item presentation and response submission. Response time is used as a proxy for cognitive effort, fluency, and guessing behavior.

Item Difficulty

Item difficulty is represented as a categorical or ordinal variable (e.g., easy, medium, hard) assigned during item design and refined during pilot calibration. Difficulty information allows the models to contextualize correctness outcomes.

Attempt Number and History

Attempt-related variables capture whether a question is being attempted for the first time or as a repeat attempt. These variables support the modeling of learning progression and practice effects.

Sequence Features

Sequence features represent the order in which concepts and items are attempted. These features help identify prerequisite violations and non-linear learning patterns.

Learner Background Covariates

Optional background variables such as academic program, prior coursework, and self-reported proficiency are incorporated to improve personalization while being handled cautiously to avoid bias.

3.6.3 Control Variables

Control variables are included to isolate the effect of the intervention from confounding factors. These include the time interval between pre-test and post-test and the quantity of recommended learning content consumed.

3.7 Data Sources and Dataset Plan

The robustness of any data-driven system is fundamentally dependent on the quality, structure, and diversity of its underlying dataset. Accordingly, this study adopts a multi-source data strategy encompassing primary data collection, secondary data utilization, and controlled synthetic data augmentation.

3.7.1 Primary Data Sources

Primary data is collected directly through the custom-built MCQ assessment platform developed as part of this research. Each learner interaction is logged in a structured format, capturing a rich set of attributes including pseudonymized user identifiers, question identifiers, timestamps, selected options, correctness indicators, response times, attempt numbers, optional confidence ratings, and selected target roles.

The platform is designed to ensure high-fidelity data capture, minimize logging errors, and maintain consistent schema definitions across all assessment sessions.

3.7.2 Secondary Data Sources

Secondary datasets obtained from publicly available repositories such as Kaggle and institutional learning analytics datasets are reviewed and adapted where appropriate.

These datasets are used primarily for exploratory validation, benchmarking, and sensitivity analysis rather than as primary training data.

3.7.3 Synthetic Data Augmentation

To address sparsity and imbalance in learner response patterns—particularly for advanced concepts or less frequently attempted topics—controlled synthetic data augmentation is employed. Synthetic records are generated using probabilistic models calibrated on pilot data distributions. Importantly, synthetic data is clearly labeled and excluded from final performance evaluation to prevent inflation of results.

3.7.4 Dataset Schema Design

The dataset schema is logically organized into multiple relational tables, including Users, Questions, Responses, Resources, and Recommendations. This modular design supports scalability, efficient querying, and clear separation of concerns.

3.8 Sampling Strategy

3.8.1 Target Population

The target population for this study includes final-year undergraduate and postgraduate students enrolled in computing-related programs, as well as early-career technical professionals preparing for entry-level roles in AI/ML and software development.

3.8.2 Sampling Method

A purposive sampling approach is adopted to ensure that participants possess baseline exposure to computing concepts while representing diverse academic backgrounds. Recruitment is conducted through university cohorts, online learning communities, and professional networks.

3.8.3 Stratification and Representation

Participants are stratified across academic programs (e.g., BCA, BTech, MCA, MTech) and self-reported proficiency levels. This stratification enhances representativeness and supports subgroup analysis during evaluation.

3.8.4 Sample Size Justification

Sample size considerations are informed by both modelling requirements and statistical power analysis. For educational impact evaluation, power calculations targeting an alpha level of 0.05 and statistical power of 0.8 suggest a minimum of 64 participants per experimental group. Larger sample sizes are targeted to account for attrition and incomplete data.

3.9 Instrument Development

Instrument development is a critical component of the research methodology, as the validity of all subsequent analyses depends on the quality of assessment instruments and supporting surveys.

3.9.1 MCQ Item Bank Construction

The MCQ item bank is constructed through a systematic, multi-stage process.

Topics are derived from consolidated AI/ML curricula and industry competency frameworks. For each concept, multiple items are authored across difficulty levels to capture varying depths of understanding.

Distractors are intentionally designed based on common misconceptions identified through teaching experience and literature review, thereby enhancing diagnostic sensitivity.

3.9.2 Item Tagging and Metadata

Each MCQ item is annotated with comprehensive metadata, including primary and secondary concepts, difficulty level, Bloom's taxonomy level, and prerequisite concept identifiers. Expert review by faculty and domain specialists ensures tagging

accuracy and conceptual validity.

3.9.3 Pilot Calibration

Prior to full deployment, items undergo pilot testing with a small cohort of participants. Classical Test Theory metrics such as item difficulty and discrimination indices are computed, and items failing to meet quality thresholds are revised or removed.

3.9.4 Survey Instruments

Supplementary survey instruments are developed to capture learner background, self-efficacy, satisfaction, perceived usefulness, and engagement with recommended resources. Survey items use validated Likert-scale formats where applicable.

3.10 Data Collection Procedure

Data collection is conducted in a structured, multi-stage manner to ensure consistency, data quality, and experimental control.

3.10.1 Platform Deployment

The assessment platform is deployed on a web-based interface connected to a relational database backend. Automated logging mechanisms capture all learner interactions with minimal manual intervention.

3.10.2 Pilot Study

A pilot study involving a limited number of participants is conducted to validate system functionality, item clarity, timing constraints, and logging accuracy. Findings from the pilot inform refinements prior to full-scale deployment.

3.10.3 Main Study Execution

In the main study, participants complete a diagnostic pre-test aligned with selected concepts or roles. Following the pre-test, participants in the treatment group receive prerequisite-aware recommendations, while control group participants receive generic learning guidance. After a defined intervention window, participants complete a post-test.

3.10.4 Resource Consumption Tracking

Where feasible, engagement with recommended resources is tracked through click logs and supplemented by self-reported measures of time spent and perceived usefulness.

3.11 Data Preparation and Quality Assurance

Data preparation constitutes a critical stage in the research methodology, as the validity, reliability, and interpretability of all subsequent modelling and evaluation outcomes are contingent upon the quality of the underlying data. In learning analytics and educational data mining contexts, raw learner interaction data is often noisy, incomplete, and heterogeneous. Accordingly, this study adopts a systematic and conservative data preparation strategy designed to preserve meaningful signal while minimizing bias and analytical distortion.

3.11.1 Data Cleaning Procedures

The initial step in data preparation involves comprehensive data cleaning. All raw logs extracted from the assessment platform are examined for duplicate records, corrupted entries, and inconsistencies arising from network interruptions or partial submissions. Duplicate response records, identified through combinations of user identifiers, question identifiers, and timestamps, are removed to prevent artificial inflation of attempt counts.

Timestamps are normalized to a consistent time zone and format to enable accurate computation of response durations and temporal trends. Response time values are recalculated where necessary to ensure alignment between recorded start and submission events. Questions skipped by learners are explicitly marked as omitted rather than treated as incorrect responses, thereby preserving semantic distinction in the dataset.

3.11.2 Handling Missing and Incomplete Data

Missing values are handled in a manner consistent with their semantic meaning. Optional confidence ratings, when absent, are retained as null values rather than imputed, as imputation could introduce artificial certainty. Background survey variables are used selectively and excluded from modelling if missingness exceeds acceptable thresholds.

3.11.3 Outlier Detection and Treatment

Response time outliers are identified using distributional analysis. Extremely long response times, defined as values exceeding the mean plus four standard deviations, are flagged for further inspection. Such outliers may indicate distraction, technical issues, or disengagement. Depending on context, these values are either capped to a maximum threshold or excluded from specific behavioural analyses while retaining correctness information.

3.11.4 Feature Engineering

Feature engineering transforms raw response logs into analytically meaningful representations. Several classes of features are derived:

Accuracy Aggregates: Per-concept accuracy rates are computed as the ratio of correct responses to total attempts for each concept.

Temporal Features: Trends in correctness across successive attempts are modeled using slope estimates to capture learning progression.

Behavioral Features: Average response time per concept, response time variance, and the proportion of rapid incorrect responses are calculated to identify guessing behavior.

Sequence-Based Features: The frequency of prerequisite violations—instances where advanced concepts are attempted successfully while foundational prerequisites remain unmastered—is quantified.

Stability Metrics: Consistency indices are computed to measure variability in learner performance across repeated attempts of the same concept.

3.11.5 Data Splitting and Leakage Prevention

To ensure robust generalization assessment, all modelling experiments employ user-level data splits. Learners appearing in the training set are entirely excluded from validation and test sets. This approach prevents information leakage and more accurately reflects real-world deployment scenarios involving previously unseen users.

3.12 Modelling Strategy

The modelling strategy adopted in this research is guided by three overarching principles: interpretability, robustness, and pedagogical alignment. While complex deep learning models may offer marginal performance gains, their opacity and data requirements render them unsuitable for the formative educational context of this study.

3.12.1 Modelling Objectives

The primary modelling objectives are to estimate per-concept mastery probabilities, compute a composite Skill Gap Index, and generate interpretable signals that can be translated into actionable learning recommendations.

3.12.2 Baseline Models

Baseline models employ simple rule-based metrics, such as percentage-correct per concept. These models provide transparency and serve as reference points against which more sophisticated approaches are evaluated.

3.12.3 Probabilistic and IRT-Inspired Models

Probabilistic models incorporating item difficulty metadata are used to derive scaled mastery estimates. These models draw inspiration from Item Response Theory (IRT) without fully adopting its parametric assumptions, thereby balancing theoretical grounding with practical flexibility.

3.12.4 Machine Learning Models

Tree-based ensemble models, including Random Forests and Gradient Boosting Machines, are employed due to their ability to handle heterogeneous features and non-linear interactions. Logistic regression models with calibrated probabilities are also used to support interpretability and comparative analysis.

3.12.5 Explainability and Model Transparency

Model explainability is prioritized through feature importance analysis and local explanation techniques. Explanatory outputs articulate how factors such as prerequisite weaknesses, response time patterns, and performance variability contribute to mastery estimates.

3.12.6 Model Evaluation Metrics

Model performance is evaluated using classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. For continuous mastery estimates, RMSE

and MAE are reported. Calibration is assessed using Brier scores and calibration plots.

3.13 Recommendation Engine Design

The recommendation engine constitutes the pedagogical core of the system. Its design is grounded in learning science principles, particularly mastery learning, scaffolding, and prerequisite alignment.

3.13.1 Knowledge Graph Representation

A directed knowledge graph encodes concepts as nodes and prerequisite relationships as edges. This representation enables systematic traversal of foundational dependencies.

3.13.2 Recommendation Generation Process

The recommendation process involves identifying the largest mastery gaps, mapping prerequisite chains, selecting appropriate learning resources, and sequencing recommendations into manageable learning plans.

3.13.3 Resource Selection and Ranking

Learning resources are selected from freely available repositories and ranked based on pedagogical clarity, duration, accessibility, and alignment with learner needs.

Micro-learning resources are prioritized for initial remediation.

3.13.4 Explainability and Feedback Loop

Each recommendation includes a concise rationale explaining why it was suggested.

Post-recommendation reassessment enables dynamic updating of mastery estimates and recommendation refinement.

3.14 Experimental Design and Evaluation Plan

A randomized, two-arm controlled experimental design is employed to evaluate educational impact. Participants are assigned to treatment and control groups, complete pre-tests and post-tests, and are compared using appropriate statistical analyses.

3.14.1 Statistical Analysis

Paired and independent statistical tests are used to assess within-subject and between-group differences. Effect sizes, confidence intervals, and adjustments for multiple comparisons are reported.

3.14.2 Robustness and Subgroup Analysis

Subgroup analyses examine differential effects by prior proficiency and academic background. Per-protocol and intention-to-treat analyses are conducted to assess robustness.

3.15 Reliability, Validity, and Trustworthiness

Content validity is ensured through expert review and curriculum mapping. Construct validity is examined through pilot statistics and external proxies. Reliability is assessed using internal consistency and test-retest measures.

3.16 Ethical Considerations and Data Governance

Ethical safeguards include informed consent, anonymization, secure data storage, fairness monitoring, and academic integrity measures. The system is explicitly positioned as a formative diagnostic tool.

3.17 Implementation Plan and Resources

The project utilizes academic supervision, domain expertise, open-source technologies, and a structured six-month development timeline encompassing taxonomy design, data collection, modelling, evaluation, and reporting.

3.18 Risk Management

Risks related to participation, data quality, model overfitting, and privacy are identified and mitigated through diversified recruitment, pilot testing, cross-validation, and strict anonymization protocols.

3.19 Expected Outputs

The expected outputs include a validated MCQ item bank, a curated dataset, trained and evaluated models, a functional recommendation engine, analytical dashboards, and empirical evidence of educational impact.

3.20 Summary

This chapter has presented and extensively elaborated and methodologically rigorous account of the research design and procedures underlying the AI-Based Skill Gap Analysis and Learning Recommendation System. By integrating robust system engineering with empirical educational evaluation, the methodology establishes a strong foundation for the analysis and results presented in subsequent chapters.

CHAPTER 4

DATA ANALYSIS, RESULTS, AND INTERPRETATION

4.1 Introduction

This chapter presents a detailed and systematic account of the dataset construction process, exploratory data analysis, modelling experiments, and quantitative results obtained from the implementation of the AI-Based Skill Gap Analysis and Learning Recommendation System. While the previous chapter focused on methodological design and research planning, the present chapter operationalizes those plans by documenting how data was instantiated, analyzed, and transformed into actionable insights. The chapter serves as the empirical core of the dissertation, linking theoretical assumptions and methodological decisions to measurable outcomes.

The objectives of this chapter are fourfold. First, it describes how the raw learner interaction data was collected, structured, and finalized into an analysis-ready dataset. Second, it presents exploratory analyses conducted to understand learner behavior, item characteristics, and concept-level performance distributions. Third, it documents the modelling experiments performed to estimate mastery probabilities and skill gaps, including comparative analysis of baseline and advanced models. Finally, it reports quantitative results related to model performance, calibration, and educational impact, thereby providing evidence for the effectiveness of the proposed system.

This chapter is written with an emphasis on clarity, traceability, and academic rigor. All analyses are aligned with the research questions and evaluation criteria defined in Chapter 3, and results are presented in a manner suitable for replication and critical assessment.

4.2 Dataset Construction Overview

Dataset construction represents a foundational step in the empirical workflow of this research. The quality of modelling outcomes and the validity of conclusions drawn are directly dependent on the integrity, representativeness, and structure of the dataset. Accordingly, dataset construction was approached as a deliberate and multi-stage process rather than a simple aggregation of raw logs.

The final dataset integrates learner assessment responses, item-level metadata, concept taxonomy information, and recommendation interaction logs. Each data source is harmonized into a unified schema that supports both descriptive analysis and predictive modelling.

4.3 Data Collection Summary

4.3.1 Source of Data

The primary source of data for this study is the custom-built MCQ assessment platform developed as part of the research. Participants interacted with the platform by selecting a target role or concept focus, completing diagnostic assessments, and—depending on experimental assignment—engaging with recommended learning resources before completing a post-assessment.

Data collection was conducted in controlled phases, beginning with a pilot deployment and followed by a main study rollout. All learner interactions were logged automatically to minimize manual intervention and reduce the risk of recording errors.

4.3.2 Nature of Collected Data

The collected data consists of multiple interconnected components:

- Learner profile data (pseudonymized), including academic background and self-reported proficiency.
- MCQ response data, capturing correctness, response time, attempt sequence, and confidence ratings.
- Item metadata, including concept tags, difficulty levels, Bloom’s taxonomy levels, and prerequisite mappings.
- Recommendation logs, indicating which learning resources were suggested and whether they were accessed.

Together, these components form a rich, multi-dimensional dataset suitable for both behavioral analysis and predictive modelling.

4.4 Final Dataset Schema and Structure

The finalized dataset is organized into a relational structure comprising multiple tables, each serving a distinct analytical purpose. This modular organization enhances scalability, maintainability, and analytical clarity.

4.4.1 Users Table

The Users table contains pseudonymized learner identifiers and optional background attributes. Personally identifiable information is excluded to preserve anonymity.

Background attributes are stored separately from performance data to reduce bias risk.

4.4.2 Questions Table

The Questions table stores item-level metadata, including question identifiers, primary and secondary concept tags, difficulty levels, Bloom’s taxonomy

classifications, and prerequisite concept identifiers. This table serves as the conceptual backbone for diagnostic inference.

4.4.3 Responses Table

The Responses table represents the core transactional dataset. Each record corresponds to a learner's interaction with a specific question and includes timestamps, selected options, correctness indicators, response times, attempt numbers, and confidence ratings.

4.4.4 Resources and Recommendations Tables

The Resources table catalogs learning materials used by the recommendation engine, while the Recommendations table logs which resources were suggested to which learners, in what sequence, and at what time.

4.5 Dataset Size and Composition

Following data cleaning and validation, the final dataset consists of several hundred unique learners and tens of thousands of individual response records. The dataset spans a wide range of concepts and difficulty levels, providing sufficient variability for modelling and evaluation.

Descriptive statistics indicate balanced representation across foundational and intermediate concepts, with controlled sparsity in advanced topics addressed through synthetic augmentation as described in Chapter 3.

4.6 Data Validation and Integrity Checks

Prior to analysis, multiple validation checks were conducted to ensure dataset integrity. These include referential integrity checks across tables, validation of

timestamp ordering, verification of concept tag consistency, and inspection of response time distributions. Records failing integrity checks were excluded from analysis.

4.7 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to gain an initial understanding of learner behavior, item characteristics, and concept-level performance trends. EDA also informed feature engineering decisions and model selection.

4.7.1 Learner Participation and Attempt Patterns

Analysis of participation patterns reveals variability in assessment engagement, with most learners attempting between a moderate number of questions per session.

Attempt distributions indicate that repeat attempts are more common for foundational concepts, suggesting iterative learning behavior.

4.7.2 Response Time Analysis

Response time distributions exhibit right-skewed characteristics typical of cognitive task data. Short response times combined with incorrect answers suggest guessing behavior, while longer response times are associated with higher difficulty items.

4.7.3 Item Difficulty and Discrimination

Item-level analysis confirms alignment between designed difficulty labels and observed correctness rates. Items with poor discrimination indices were flagged during pilot analysis and are underrepresented in the final dataset.

4.7.4 Concept-Level Performance Distribution

Concept-level aggregation reveals substantial variability in mastery across learners. Foundational concepts show higher average mastery, while advanced topics exhibit greater dispersion, underscoring the need for personalized remediation.

4.8 Extended Exploratory Data Analysis

This section presents an in-depth exploratory data analysis (EDA) of the finalized dataset with the objective of uncovering structural patterns, behavioral trends, and latent relationships that inform subsequent modelling decisions. EDA serves not merely as a descriptive exercise but as an analytical bridge between raw data and predictive modelling. All analyses presented here are conducted on the cleaned, validated dataset described in the previous section.

4.8.1 Distribution of Learner Engagement

Learner engagement is first examined through the distribution of assessment attempts per participant. Analysis reveals a right-skewed distribution, with a majority of learners attempting a moderate number of questions and a smaller subset exhibiting very high engagement levels. This pattern is consistent with voluntary learning environments, where intrinsic motivation and time availability vary significantly across individuals.

Session-level analysis indicates that learners tend to complete assessments in multiple short sessions rather than a single prolonged session. This observation supports the decision to incorporate temporal and attempt-based features into the modelling pipeline, as learning behavior unfolds over time rather than instantaneously.

4.8.2 Concept Coverage and Attempt Frequency

The frequency of attempts across concepts is analyzed to assess coverage balance. Foundational concepts such as programming basics, data preprocessing, and descriptive statistics exhibit higher attempt frequencies, while advanced topics such as optimization techniques and model tuning show lower engagement. This imbalance reflects both curriculum sequencing and learner confidence levels. The observed imbalance further justifies the controlled use of synthetic data augmentation for underrepresented concepts, as discussed in Chapter 3. Importantly, augmentation is applied conservatively to avoid distorting authentic learning patterns.

4.8.3 Correctness Rate Analysis

Aggregate correctness rates are computed at item, concept, and difficulty levels. As expected, correctness decreases monotonically with increasing item difficulty.

However, substantial variance is observed within difficulty categories, indicating that difficulty labels alone are insufficient to capture item challenge.

Concept-level correctness analysis reveals that learners often perform inconsistently across related concepts, reinforcing the need for fine-grained mastery estimation rather than aggregate scoring. For example, learners may demonstrate adequate understanding of basic probability while struggling with conditional probability, despite both being categorized under the same broad domain.

4.8.4 Response Time Behavior

Response time distributions are analyzed to gain insight into cognitive processing and engagement. The distributions are positively skewed, with median response times varying significantly by concept and difficulty. Foundational concepts generally exhibit shorter response times, suggesting fluency, while advanced concepts are associated with longer deliberation.

Joint analysis of response time and correctness reveals distinct behavioral clusters. Short response times combined with incorrect answers are indicative of guessing or low engagement, whereas longer response times with correct answers suggest deliberate reasoning. These observations motivate the inclusion of interaction features between response time and correctness in the modelling phase.

4.8.5 Attempt Sequence and Learning Progression

Sequential analysis examines how learner performance evolves across repeated attempts. Many learners demonstrate improvement over successive attempts on the same concept, reflecting learning and practice effects. However, improvement trajectories vary widely, with some learners exhibiting plateauing or fluctuating performance.

Sequence analysis also reveals instances of prerequisite violations, where learners correctly answer advanced items while failing prerequisite concepts. Such patterns highlight superficial or memorized understanding and underscore the pedagogical importance of prerequisite-aware remediation.

4.8.6 Variability and Stability of Performance

Stability metrics are computed to assess consistency of learner performance across attempts. High variability is observed among learners with lower overall mastery, whereas high-mastery learners tend to exhibit stable performance. These findings suggest that variability itself is an informative signal for mastery estimation and risk identification.

4.9 Correlation and Feature Relationship Analysis

Correlation analysis is conducted to examine relationships among engineered features and outcome variables. While correctness and mastery probability are positively

correlated, response time exhibits a non-linear relationship with performance.

Moderate response times are often associated with optimal performance, whereas extremely short or long times correlate with lower accuracy.

Background variables such as prior coursework show modest correlation with baseline mastery but weaker association with learning gains, indicating that prior exposure does not guarantee effective learning progression. This finding supports the system’s emphasis on individualized diagnostics rather than reliance on background credentials.

4.10 Insights Derived from Exploratory Analysis

The EDA yields several key insights that directly inform modelling decisions. First, learner behavior is heterogeneous and temporally dynamic, necessitating models that incorporate behavioral and sequence-based features. Second, aggregate scores mask important conceptual weaknesses, reinforcing the value of per-concept mastery estimation. Third, prerequisite violations are common and pedagogically significant, justifying the design of a prerequisite-aware recommendation engine.

4.11 Transition to Modelling Experiments

Based on the exploratory findings, the modelling phase focuses on approaches capable of handling heterogeneous features, capturing non-linear relationships, and producing interpretable outputs. The next section documents the modelling experiments conducted, beginning with baseline approaches and progressing to more advanced machine learning models.

4.12 Modelling Experiments and Analytical Framework

This section documents the modelling experiments conducted to estimate learner mastery, identify skill gaps, and evaluate the predictive performance of alternative modelling approaches. All experiments are executed in alignment with the modelling strategy articulated in Chapter 3 and are designed to balance predictive accuracy with interpretability and educational relevance.

4.12.1 Experimental Setup and Data Splits

All modelling experiments employ user-level data partitioning to prevent information leakage and to simulate real-world deployment scenarios involving previously unseen learners. The dataset is divided into training, validation, and test sets using stratified sampling to preserve concept-level representation. Hyperparameter tuning is conducted exclusively on the training and validation sets, with final performance reported on the held-out test set.

To ensure robustness, multiple random seeds are used, and reported results represent averages across runs where applicable. Synthetic data, where used for balancing during training, is excluded from final evaluation metrics.

4.12.2 Baseline Rule-Based Models

Baseline models are implemented to establish reference performance levels and to contextualize the gains achieved by more advanced approaches. The primary baseline computes per-concept mastery as the percentage of correctly answered items for each learner.

While this approach offers maximal transparency, it is sensitive to small sample sizes and fails to account for item difficulty or behavioral signals. Baseline results demonstrate reasonable performance for well-sampled concepts but poor stability for sparsely attempted topics, highlighting the need for more sophisticated modelling.

4.12.3 Probabilistic and IRT-Inspired Models

Probabilistic models incorporating item difficulty metadata are developed to address limitations of the baseline. These models weight learner responses by item difficulty and estimate mastery probabilities using logistic transformations.

Results indicate improved stability across concepts with varying question counts.

Calibration analysis shows closer alignment between predicted mastery probabilities and observed correctness frequencies compared to the baseline model. These findings support the inclusion of difficulty-aware probabilistic modelling as an intermediate step between simple heuristics and machine learning.

4.12.4 Logistic Regression Models

Logistic regression models are trained using engineered features including correctness aggregates, response time statistics, variability measures, and prerequisite violation counts. Probability calibration techniques are applied to ensure interpretability.

Logistic models achieve improved discrimination performance relative to probabilistic baselines while maintaining interpretability through coefficient analysis. Feature coefficients reveal that prerequisite violations and response time variability are significant negative predictors of mastery.

4.12.5 Tree-Based Ensemble Models

Tree-based ensemble models, including Random Forests and Gradient Boosting Machines, are trained to capture non-linear interactions among heterogeneous features. These models demonstrate the highest predictive performance across most evaluation metrics.

Feature importance analysis indicates that correctness-based aggregates, prerequisite violation frequency, and temporal performance trends are among the most influential

predictors. Despite their superior accuracy, ensemble models are accompanied by interpretability mechanisms to support educational transparency.

4.13 Model Performance Evaluation

4.13.1 Classification Metrics

Model performance is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC for binary mastery classification derived by thresholding mastery probabilities. Ensemble models consistently outperform baselines, with statistically significant improvements in ROC-AUC.

4.13.2 Regression Metrics

For continuous mastery estimation, RMSE and MAE are computed. Probabilistic and ensemble models exhibit lower error values, indicating more precise mastery estimation.

4.13.3 Calibration Analysis

Calibration plots and Brier scores are used to assess probabilistic reliability. Difficulty-aware models demonstrate superior calibration, reducing overconfidence observed in baseline approaches.

4.14 Skill Gap Index Analysis

The Skill Gap Index is analysed to assess its distribution, sensitivity, and interpretability. Results show that the index effectively differentiates learners across proficiency levels and aligns with external indicators such as academic background and self-reported confidence.

Role-level aggregation of the Skill Gap Index reveals distinct readiness profiles, supporting its utility as a role-aligned diagnostic metric.

4.15 Interpretation of Modelling Results

The modelling results demonstrate that incorporating behavioral and sequence-based features yields meaningful gains in diagnostic accuracy. Importantly, interpretability analyses confirm that model outputs can be translated into actionable pedagogical insights, validating the design philosophy of the system.

4.16 Transition to Educational Impact Analysis

Having established the predictive validity of the modelling approaches, the next section evaluates the educational impact of using these diagnostics to drive prerequisite-aware learning recommendations.

4.17 Educational Impact Analysis

This section evaluates the educational effectiveness of the AI-Based Skill Gap Analysis and Learning Recommendation System by examining learner outcomes following the intervention. While prior sections focused on diagnostic accuracy and predictive validity, the present analysis addresses the core pedagogical question of whether prerequisite-aware, personalized recommendations lead to measurable improvements in learner mastery compared to generic study guidance.

4.17.1 Experimental Groups and Intervention Overview

Participants included in the educational impact analysis were those who completed both the pre-test and post-test within the defined intervention window. Learners

were randomly assigned to one of two groups: a treatment group that received prerequisite-aware, personalized learning recommendations generated by the system, and a control group that received non-personalized, flat lists of learning resources or general study instructions.

Both groups were provided with an equivalent amount of study time and access to free learning resources. The distinguishing factor between groups was the structure, sequencing, and explanatory framing of the recommendations.

4.17.2 Pre-Test Baseline Equivalence

Baseline equivalence between treatment and control groups was assessed using pre-test mastery probabilities and aggregate correctness scores. Statistical analysis confirmed no significant differences between groups at baseline, indicating that randomization was effective and that subsequent differences could be attributed to the intervention rather than pre-existing disparities.

4.17.3 Pre-Test and Post-Test Mastery Gains

Within-group comparisons reveal that learners in the treatment group exhibited substantial improvements in concept mastery probabilities following the intervention. Average mastery gains were observed across foundational, intermediate, and advanced concepts, with particularly strong improvements in concepts previously identified as prerequisites.

In contrast, the control group demonstrated modest improvements, primarily concentrated in concepts that were already partially mastered at baseline. This pattern suggests that unguided or generic study tends to reinforce existing knowledge rather than address foundational gaps.

4.17.4 Statistical Significance Testing

Paired statistical tests were conducted to evaluate within-group pre-test and post-test differences. For normally distributed mastery gains, paired t-tests were applied, while non-parametric alternatives were used where distributional assumptions were violated. Results indicate statistically significant improvements in the treatment group across most assessed concepts.

Between-group comparisons were conducted using independent samples tests.

Treatment group gains were significantly higher than control group gains, with medium to large effect sizes observed for key foundational and intermediate concepts.

4.17.5 Effect Size and Practical Significance

Effect sizes were calculated using Cohen's d to assess practical significance. Results indicate medium effect sizes for overall mastery improvement and large effect sizes for prerequisite-heavy concept clusters. These findings underscore the educational value of prerequisite-aware recommendation sequencing.

4.18 Role-Level Readiness Improvement

Role-level readiness was evaluated by aggregating concept mastery probabilities according to role-specific competency profiles. Learners in the treatment group exhibited significant reductions in Skill Gap Index values across all evaluated roles, indicating improved alignment with role expectations.

Notably, the largest readiness gains were observed for roles with strong dependency chains, such as Machine Learning Engineer and Data Analyst, further validating the system's prerequisite modeling approach.

4.19 Learner Engagement and Resource Utilization

Analysis of recommendation interaction logs reveals higher engagement levels among treatment group learners. Treatment participants accessed a greater proportion of recommended resources and reported higher perceived usefulness. Correlation analysis indicates a positive relationship between resource engagement and mastery gains, particularly for structured learning sequences.

4.20 Learner Satisfaction and Usability Feedback

Post-intervention surveys were analyzed to assess learner perceptions of system usability, clarity, and trustworthiness. Treatment group learners reported higher satisfaction scores, citing clear explanations, logical sequencing, and targeted content as key strengths.

Qualitative feedback suggests that learners valued understanding why specific topics were recommended, which increased motivation and adherence. Control group feedback frequently indicated uncertainty about what to study next.

4.21 Discussion of Findings

The findings presented in this chapter provide strong empirical support for the effectiveness of the proposed system. The results demonstrate that accurate diagnostic modelling, when coupled with pedagogically informed recommendation strategies, can produce meaningful improvements in learner outcomes.

Importantly, the study highlights that personalization alone is insufficient; the structure and sequencing of recommendations based on prerequisite knowledge play

a critical role in driving learning gains. These insights align with established theories of mastery learning and scaffolding.

4.22 Limitations of the Empirical Study

Despite positive findings, certain limitations must be acknowledged. The reliance on MCQ-based assessments constrains the scope of evaluated skills. Additionally, intervention duration was limited, and long-term retention effects were not measured. These limitations suggest directions for future research rather than detracting from the validity of the current findings.

4.23 Chapter Summary

This chapter presented a comprehensive empirical evaluation of the AI-Based Skill Gap Analysis and Learning Recommendation System. Through detailed dataset construction, exploratory analysis, modelling experiments, and educational impact assessment, the chapter demonstrated both the predictive validity and pedagogical effectiveness of the proposed approach. The results confirm that prerequisite-aware, personalized recommendations significantly outperform generic study guidance in improving learner mastery and role readiness. The following chapter will synthesize these findings, discuss broader implications, and outline future research directions.

CHAPTER 5

DISCUSSION, CONCLUSION AND FUTURE SCOPE

5.1 Introduction

This chapter synthesizes the findings presented in the preceding chapters and situates them within the broader academic, technical, and educational context. While Chapters 3 and 4 focused on methodology and empirical results respectively, the present chapter interprets those results, discusses their implications, acknowledges limitations, and outlines directions for future research and system enhancement. The chapter serves as the intellectual closure of the dissertation by connecting research objectives to outcomes and demonstrating the contribution of the study to the domains of learning analytics, educational data mining, and AI-driven personalized learning systems.

The discussion presented in this chapter is structured around the original research objectives and questions. Each major finding is critically examined in light of existing literature and established theories of learning, assessment, and personalization. The chapter also reflects on the practical relevance of the proposed system for learners, educators, and institutions, particularly in the context of employability-focused technical education.

5.2 Summary of the Research Work

This research set out to design, implement, and evaluate an AI-Based Skill Gap Analysis and Learning Recommendation System capable of diagnosing fine-grained concept-level and role-level skill gaps using MCQ assessment data. The study was

motivated by the persistent mismatch between academic learning outcomes and industry expectations in the rapidly evolving AI/ML domain.

The work began with the identification of limitations in traditional assessment systems, particularly their inability to provide actionable diagnostic insights. In response, the research proposed a structured concept taxonomy with explicit prerequisite relationships, enabling assessment data to be interpreted at a granular level. A comprehensive MCQ item bank was developed and tagged to this taxonomy, forming the basis for data collection.

A custom assessment platform was engineered to collect high-quality learner interaction data, including behavioral metadata such as response time and attempt history. This data was processed through a carefully designed modelling pipeline to estimate per-concept mastery probabilities and compute a standardized Skill Gap Index. A knowledge-graph-driven recommendation engine was then used to transform diagnostic outputs into personalized, prerequisite-aware learning pathways using freely available educational resources.

Empirical evaluation through controlled experiments demonstrated that learners receiving structured, prerequisite-aware recommendations achieved significantly greater mastery gains than those receiving generic study guidance. These outcomes collectively address the research objectives and validate the effectiveness of the proposed approach.

5.3 Findings of the Study

The analysis of the experimental results clearly shows that learners who used the AI-Based Skill Gap Analysis and Learning Recommendation System performed significantly better than those who followed generic study guidance. Learners in the

treatment group demonstrated a much higher improvement in concept mastery after the intervention. On average, their Skill Gap Index decreased substantially across all major AI and machine learning topics, indicating stronger readiness for role-based competencies. In contrast, learners in the control group showed only limited improvement, mainly in topics they already partially understood.

One of the most important findings is that prerequisite-aware learning sequences produced superior learning outcomes. Learners who were guided to first study foundational concepts before moving to advanced topics achieved larger mastery gains than those who were given unstructured or flat learning material. The highest improvements were observed in roles such as Machine Learning Engineer and Data Analyst, where concepts are strongly interdependent. This proves that respecting prerequisite relationships leads to deeper and more stable learning.

Another significant finding is that fine-grained concept-level diagnosis is far more effective than traditional overall scoring. Many learners who appeared competent based on total test scores were found to have serious gaps in foundational topics. The AI-based mastery models successfully detected these hidden weaknesses and allowed the system to recommend targeted learning resources. As a result, learners were able to focus their efforts on exactly the areas that needed improvement, rather than wasting time on already mastered topics.

The study also found that learner engagement and satisfaction were higher when recommendations were explained. Students who received clear reasons for why certain topics were recommended reported greater trust in the system and followed the learning pathways more consistently. This increased engagement directly contributed to better learning outcomes, confirming the importance of explainable AI in education.

Overall, the findings confirm that the proposed system is more effective, more accurate, and more educationally valuable than traditional assessment and learning methods. It successfully reduces skill gaps, improves role readiness, and provides learners with clear, structured guidance on what to study next.

5.4 Comparison with Existing Literature and Systems

The findings of this study align with and extend prior research in learning analytics and adaptive learning systems. Existing literature has demonstrated the limitations of aggregate assessment scores and the potential of mastery-based approaches. However, many prior systems rely on opaque models or proprietary datasets.

In contrast, the present study emphasizes interpretability, prerequisite modeling, and the use of freely available resources. The demonstrated educational impact of prerequisite-aware sequencing contributes empirical evidence to theoretical claims regarding scaffolding and mastery learning.

5.5 Practical Implications

5.5.1 Implications for Learners

For learners, the system provides clear diagnostic feedback and structured guidance, reducing uncertainty about what to study next. The emphasis on foundational gaps supports more efficient and confidence-building learning trajectories.

5.5.2 Implications for Educators

Educators can use aggregated diagnostics to identify common conceptual bottlenecks and to tailor instruction accordingly. The system's interpretability supports informed pedagogical intervention rather than automated decision-making.

5.5.3 Implications for Institutions

Institutions may leverage such systems to enhance curriculum alignment with industry needs, support personalized learning at scale, and improve employability outcomes without reliance on costly proprietary platforms.

5.6 Limitations of the Study

While the findings of this research demonstrate the effectiveness and practical value of the proposed AI-Based Skill Gap Analysis and Learning Recommendation System, it is essential to acknowledge the limitations inherent in the study. A transparent discussion of limitations strengthens the credibility of the research and provides appropriate context for interpreting the results.

5.6.1 Assessment Format Constraints

One of the primary limitations of the study is the exclusive reliance on multiple-choice questions as the assessment modality. Although MCQs offer scalability, objectivity, and ease of automated analysis, they are inherently limited in their ability to capture complex problem-solving skills, creative reasoning, and hands-on programming competence. Certain higher-order cognitive skills, such as algorithm design and debugging proficiency, cannot be fully assessed through MCQs alone.

5.6.2 Duration of Intervention

The intervention window used for evaluating educational impact was relatively short. While significant mastery gains were observed within this timeframe, the study does not evaluate long-term retention or transfer of learning. Consequently, the durability of observed improvements beyond the post-test period remains an open question.

5.6.3 Sample Characteristics

The participant pool, while diverse in academic background, was limited primarily to students and early-career professionals within computing disciplines. As a result, the generalizability of findings to non-technical learners or to more experienced professionals may be constrained. Additionally, participation was voluntary, which may introduce self-selection bias favoring motivated learners.

5.6.4 Modelling Assumptions

Although interpretability was prioritized, the modelling approaches employed still involve assumptions regarding feature relevance and learner behavior. For example, response time is treated as a proxy for cognitive effort, which may not hold uniformly across all learners or contexts. These assumptions, while grounded in prior research, introduce potential sources of error.

5.6.5 Resource Quality Variability

The recommendation engine relies on freely available learning resources, whose quality and pedagogical effectiveness can vary. While resources were curated and ranked, differences in presentation style and depth may influence individual learner engagement and outcomes.

5.7 Future Scope and Research Directions

The limitations identified above suggest several promising directions for future research and system enhancement. These directions span technical, pedagogical, and deployment-related dimensions.

5.7.1 Expansion of Assessment Modalities

Future iterations of the system could integrate additional assessment formats, such as coding exercises, short-answer questions, and project-based tasks. Incorporating

automated code evaluation and rubric-based scoring would enable richer skill diagnostics and broader competency coverage.

5.7.2 Advanced Modelling Techniques

While this study deliberately prioritized interpretability, future research could explore hybrid approaches combining interpretable models with deep learning techniques for sequence modelling. Recurrent or transformer-based models may capture temporal learning dynamics more effectively, provided appropriate explainability mechanisms are incorporated.

5.7.3 Adaptive and Real-Time Personalization

The recommendation engine could be extended to support real-time adaptation, dynamically adjusting learning paths based on ongoing learner interactions. Such extensions would enable more responsive personalization and continuous feedback loops.

5.7.4 Longitudinal and Large-Scale Studies

Future studies could examine the long-term impact of prerequisite-aware recommendations on knowledge retention, skill transfer, and career outcomes. Large-scale deployments across multiple institutions would enable more robust statistical analysis and subgroup comparisons.

5.7.5 Integration with Learning Management Systems

Integrating the proposed system with existing Learning Management Systems (LMS) would facilitate seamless adoption in academic environments. Such integration could support instructor dashboards, curriculum analytics, and cohort-level interventions.

5.7.6 Ethical and Fairness Enhancements

Future work may further investigate fairness-aware modelling techniques to proactively mitigate potential biases. Continuous monitoring and participatory design approaches involving diverse learner populations could enhance ethical robustness.

5.8 Final Conclusions

This dissertation set out to address a critical and persistent challenge in contemporary technical education: the gap between assessed academic performance and demonstrable, role-aligned skill readiness in the field of Artificial Intelligence and Machine Learning. Through the design, implementation, and empirical evaluation of an AI-Based Skill Gap Analysis and Learning Recommendation System, the research demonstrates that it is both feasible and educationally valuable to move beyond aggregate scoring toward fine-grained, concept-level diagnostics coupled with prerequisite-aware personalization.

The findings presented across Chapters 3 and 4 confirm that learner performance data—when enriched with behavioral metadata and interpreted through a structured knowledge taxonomy—can yield reliable estimates of conceptual mastery. These estimates, when operationalized through interpretable models and transformed into structured learning pathways, lead to statistically significant and practically meaningful improvements in learner outcomes. In particular, the study provides strong evidence that prerequisite-aware sequencing is a decisive factor in improving mastery, especially for concepts characterized by deep dependency chains.

Importantly, the research underscores that effective personalization in education is not merely a function of recommending different resources to different learners. Rather, it is the careful diagnosis of foundational gaps, the transparent explanation of learning needs, and the structured sequencing of remediation that collectively

drive learning gains. This insight has implications that extend beyond the specific system developed in this study and speak to broader principles of adaptive learning design.

5.9 Contributions of the Research

The research makes several substantive contributions to both academic inquiry and practical system design:

First, it contributes a structured methodological framework for diagnosing fine-grained skill gaps using MCQ-based assessment data augmented with behavioral features. This framework bridges psychometric principles with modern machine learning techniques while preserving interpretability.

Second, the study introduces a role-aligned Skill Gap Index that provides a standardized and interpretable measure of learner readiness relative to occupational competency profiles. This metric enables meaningful aggregation of concept-level diagnostics into actionable role-level insights.

Third, the research demonstrates the pedagogical effectiveness of a knowledge-graph-driven, prerequisite-aware recommendation engine. Empirical evidence from controlled experiments validates the superiority of structured learning pathways over generic study guidance.

Fourth, the dissertation contributes a practical, open-resource-oriented approach to personalized learning. By prioritizing freely available educational materials, the system supports scalability and equitable access, which are particularly relevant in resource-constrained educational contexts.

Finally, the research contributes empirical evidence supporting the role of explainability in AI-driven educational systems. Learner trust, engagement, and

satisfaction were demonstrably enhanced when recommendations were accompanied by clear, human-readable rationales.

5.10 Implications for Practice and Policy

From a practical perspective, the outcomes of this research suggest that institutions can leverage data-driven diagnostic systems to enhance curriculum alignment, identify systemic learning bottlenecks, and support individualized remediation at scale. Educators can use such systems as decision-support tools rather than as replacements for pedagogical judgment, enabling more targeted and informed instructional interventions.

At a policy level, the findings support the inclusion of diagnostic and formative assessment frameworks in technology education programs. Such frameworks can complement traditional summative assessments and contribute to improved employability outcomes without necessitating extensive curricular overhauls.

5.11 Closing Remarks

In conclusion, this dissertation demonstrates that AI-driven skill gap analysis and personalized recommendation systems, when designed with methodological rigor, pedagogical grounding, and ethical responsibility, can meaningfully enhance learning outcomes in complex technical domains. While the system developed in this study represents one instantiation of this approach, the underlying principles are broadly applicable across disciplines and educational contexts.

The research affirms that the future of technical education lies not in more content, but in smarter diagnosis, clearer guidance, and structured pathways that respect how learners acquire knowledge. It is hoped that the insights and contributions of this

work will inform future research, inspire practical adoption, and contribute to the ongoing evolution of intelligent, learner-centered educational systems.

CHAPTER 6

SYSTEM IMPLEMENTATION, ARCHITECTURE AND DEPLOYMENT

6.1 Introduction

This chapter presents a detailed account of the practical implementation of the AI-Based Skill Gap Analysis and Learning Recommendation System developed as part of this research. While the previous chapters focused on methodological design, empirical evaluation, and theoretical discussion, the present chapter bridges research concepts with their concrete realization in the form of a working software system. The objective of this chapter is to document how the proposed architecture, models, and processes were translated into an operational system that can be deployed, tested, and extended in real-world educational settings.

The chapter elaborates on system architecture, technology stack, database design, assessment workflow, modelling pipeline integration, recommendation engine execution, dashboard visualization, and deployment considerations. Emphasis is placed on clarity, reproducibility, and alignment with the design principles articulated in earlier chapters. The implementation details described here are sufficient to enable replication or further enhancement by future researchers or developers.

6.2 Overall System Architecture

The AI-Based Skill Gap Analysis and Learning Recommendation System is designed as a modular, layered architecture to ensure scalability, maintainability, and flexibility. The system follows a loosely coupled architecture in which each major functional component operates independently while communicating through well-defined interfaces.

At a high level, the system architecture consists of the following layers:

- Presentation Layer (User Interface)
- Application Layer (Assessment and Recommendation Logic)
- Analytics and Modelling Layer
- Data Storage Layer
- Visualization and Reporting Layer

This separation of concerns allows individual components to be modified or extended without disrupting the overall system.

6.2.1 Presentation Layer

The presentation layer provides the primary interface through which learners and administrators interact with the system. It includes web-based forms for assessment delivery, role selection, and feedback display. The interface is intentionally kept minimalistic to reduce cognitive load and to focus learner attention on assessment tasks and recommendations.

User interactions at this layer include:

- Selecting target roles or concept-focused assessments
- Attempting MCQ assessments
- Viewing personalized learning recommendations
- Accessing recommended learning resources

6.2.2 Application Layer

The application layer manages core system logic, including assessment orchestration, response validation, session management, and recommendation triggering. This layer serves as the intermediary between user-facing interfaces and backend analytics services.

Business rules such as assessment timing, randomized question selection, attempt limits, and role-specific assessment configuration are implemented at this layer.

6.3 Analytics and Modelling Pipeline Integration

The analytics and modelling layer implements the mastery estimation and skill gap computation logic described in Chapter 3. This layer processes cleaned and engineered features derived from learner response logs and applies trained models to generate per-concept mastery probabilities.

The modelling pipeline follows a structured flow:

1. Ingestion of response data from the database
2. Feature engineering and aggregation
3. Application of trained predictive models
4. Calibration and normalization of mastery probabilities
5. Computation of Skill Gap Index values

Model inference is executed asynchronously to ensure that assessment delivery remains responsive.

6.4 Recommendation Engine Execution

The recommendation engine operates as a rule-guided, data-driven service that consumes mastery estimates and outputs structured learning pathways. Using the concept knowledge graph, the engine identifies unmet prerequisites, prioritizes remediation, and selects appropriate learning resources.

Each recommendation includes:

- Concept focus

- Resource link and type
- Estimated duration
- Explanatory rationale

This information is stored in the database and displayed to learners through the presentation layer.

6.5 Database Design and Storage

The system employs a relational database to store assessment data, learner profiles, modelling outputs, and recommendation logs. The database schema mirrors the logical structure described in Chapter 3 and supports efficient querying and analytics.

Key tables include:

- Users
- Questions
- Responses
- Concepts
- Resources
- Recommendations

Normalization principles are followed to minimize redundancy while preserving query performance.

6.6 Dashboard and Visualization

To support monitoring and analysis, the system integrates a visualization layer using business intelligence tools. Dashboards display individual learner progress, cohort-level trends, concept mastery distributions, and role readiness summaries.

Visualizations are designed to be interpretable by non-technical stakeholders, including educators and administrators.

6.7 Assessment Workflow and Execution Logic

This section describes the end-to-end workflow governing how assessments are delivered, executed, and recorded within the system. The assessment workflow has been designed to ensure consistency, fairness, data integrity, and alignment with the diagnostic objectives of the research.

6.7.1 Role Selection and Assessment Configuration

At the initiation of an assessment session, learners are prompted to select a target role or opt for a concept-focused diagnostic mode. Role selection triggers dynamic configuration of the assessment, including the selection of relevant concepts, difficulty distribution, and question pool composition. This configuration logic ensures that assessments are aligned with role-specific competency profiles while maintaining coverage of foundational prerequisites.

For role-based assessments, the system constructs a balanced question set by sampling items across multiple concepts and difficulty levels. Constraints are applied to avoid over-representation of any single concept and to ensure that prerequisite concepts are adequately represented.

6.7.2 Question Randomization and Timing Controls

To minimize the risk of answer sharing and to preserve assessment integrity, questions are randomized at both the item and option levels. Each learner receives a unique sequence of questions and shuffled answer options. Timing controls are implemented to record response durations accurately and to discourage prolonged inactivity.

The system enforces soft timing constraints rather than strict time limits, recognizing that diagnostic accuracy may be compromised under excessive time pressure. However, unusually long response times are flagged for analytical review.

6.7.3 Response Capture and Validation

Upon submission of each response, the system performs validation checks to ensure data completeness and consistency. Selected options are validated against the question schema, and timestamps are recorded with millisecond precision. Responses are immediately written to the database to prevent data loss in the event of session interruption.

Optional confidence ratings are collected where enabled and stored separately to avoid influencing core modelling features unless explicitly required.

6.7.4 Session Management and Fault Tolerance

The assessment module incorporates session management mechanisms to handle interruptions due to network instability or user navigation events. Partial sessions are preserved, allowing learners to resume assessments where appropriate. Incomplete responses are clearly marked to distinguish them from incorrect attempts.

6.8 Data Logging and Event Tracking

Robust data logging is central to the system's diagnostic capabilities. Every learner interaction is logged as a discrete event, enabling granular reconstruction of assessment behavior.

6.8.1 Logging Architecture

The logging subsystem records events including session start and end, question presentation, response submission, and resource access. Logs are timestamped and associated with pseudonymized user identifiers. This event-based architecture supports both real-time analytics and retrospective analysis.

6.8.2 Data Integrity and Consistency Checks

Periodic integrity checks are performed to ensure consistency between logged events and persisted records. Referential integrity constraints in the database prevent orphaned records and ensure alignment across tables.

6.9 Security, Privacy, and Access Control

Security and privacy considerations are integral to the system's implementation, particularly given the handling of learner performance data.

6.9.1 Authentication and Authorization

Basic authentication mechanisms are implemented to restrict access to administrative interfaces. Role-based access control ensures that learners, instructors, and administrators have access only to appropriate system functions.

6.9.2 Data Privacy Measures

All user identifiers are pseudonymized prior to storage. Personally identifiable information is excluded from analytical datasets. Access to raw data is restricted to authorized personnel, and audit logs are maintained to track data access events.

6.9.3 Protection Against Misuse

Measures are implemented to mitigate potential misuse, including rate limiting, input validation, and monitoring for anomalous activity patterns. These measures support both system stability and academic integrity.

6.10 Error Handling and System Robustness

The system incorporates comprehensive error handling to ensure robustness under real-world operating conditions. Input validation errors, database connectivity issues, and model inference failures are logged with descriptive messages to facilitate debugging and maintenance.

Graceful degradation strategies are employed where possible. For example, if model inference services are temporarily unavailable, the system can fall back to baseline diagnostic logic while flagging the issue for resolution.

6.11 Deployment Architecture and Environment Setup

This section describes the deployment architecture adopted for the AI-Based Skill Gap Analysis and Learning Recommendation System, detailing how the system components are hosted, executed, and coordinated in an operational environment. Deployment decisions are guided by considerations of scalability, reliability, cost-efficiency, and ease of maintenance, consistent with the academic and practical constraints of the project.

6.11.1 Deployment Model

The system follows a centralized web-based deployment model, enabling learners to access assessments and recommendations through standard web browsers without

requiring specialized client-side installations. This model supports broad accessibility and simplifies updates and maintenance.

Core system components—including the application logic, modelling services, and database—are deployed on a server environment configured to support concurrent user access. The architecture is designed to be adaptable, allowing migration from a local or institutional server to a cloud-based infrastructure if required.

6.11.2 Environment Configuration

The development and deployment environments are logically separated to ensure system stability. The development environment is used for iterative testing, debugging, and model experimentation, while the deployment environment hosts the stable, validated system used for data collection and evaluation.

Environment configuration includes:

- Web server setup for hosting the application interface
- Database server configuration with controlled access
- Analytics runtime for executing model inference
- Scheduled jobs for maintenance and data backup

Configuration parameters such as database credentials, API endpoints, and logging levels are externalized to facilitate secure and flexible deployment.

6.11.3 Scalability Considerations

Scalability is addressed through modular system design and asynchronous processing. Assessment delivery and response logging operate independently of model inference, ensuring that temporary increases in user activity do not degrade system responsiveness.

The modelling pipeline is designed to process inference requests asynchronously, allowing batch processing of mastery estimation during peak usage periods. Database indexing and query optimization are employed to support efficient retrieval of assessment and recommendation data.

6.11.4 Performance Optimization

Performance optimization efforts focus on minimizing latency during assessment delivery and recommendation retrieval. Caching mechanisms are used to store frequently accessed metadata such as concept mappings and resource lists. Query execution plans are monitored and refined to reduce database load.

Model inference time is monitored to ensure that mastery estimation and recommendation generation occur within acceptable response windows. Where necessary, simplified fallback logic is employed to maintain user experience.

6.12 Integration Readiness and Extensibility

The system is designed with integration readiness in mind, enabling future incorporation into broader educational ecosystems. Clear interface boundaries and modular services facilitate integration with Learning Management Systems (LMS), student information systems, or external analytics platforms.

Extensibility considerations include: • Ability to add new concepts and roles without restructuring the database • Support for additional assessment types in future iterations • Modular replacement or upgrading of modelling components

These design choices ensure that the system can evolve beyond the scope of the current research project.

6.13 Implementation Constraints and Practical Challenges

Several practical constraints influenced implementation decisions. Limited computational resources necessitated the selection of efficient, interpretable models over computationally intensive deep learning approaches. Time constraints associated

with academic project timelines required prioritization of core functionalities over optional enhancements.

Data sparsity for advanced concepts posed challenges for model training, which were mitigated through controlled augmentation and conservative evaluation practices. Additionally, variability in external learning resource quality required careful curation and ranking strategies.

6.14 Lessons Learned from System Implementation

The system implementation process yielded several important lessons. First, early investment in clean data logging and schema design significantly reduced downstream complexity in modelling and evaluation. Second, interpretability-focused model selection enhanced stakeholder trust and facilitated meaningful explanation of results.

Third, prerequisite modeling emerged as a critical design element with both technical and pedagogical implications. Explicit representation of conceptual dependencies improved diagnostic accuracy and recommendation effectiveness.

Finally, iterative testing and pilot deployment were essential for identifying usability issues and refining system behavior.

6.15 Summary of Chapter 6

This chapter documented the practical realization of the AI-Based Skill Gap Analysis and Learning Recommendation System, detailing its architecture, assessment workflow, analytics integration, deployment strategy, and implementation challenges. The discussion demonstrates that the system is not merely a conceptual prototype

but a thoughtfully engineered artifact capable of supporting real-world diagnostic and learning scenarios.

The implementation choices described in this chapter align closely with the methodological principles outlined earlier in the dissertation, reinforcing the coherence between research design, empirical evaluation, and system realization. The final chapter of the dissertation consolidates the overall contributions of the research and reflects on its broader significance.

CHAPTER 7

RECOMMENDATIONS AND LIMITATIONS OF THE STUDY

7.1 Introduction

This chapter presents the practical recommendations derived from the findings of the study and provides a critical account of the limitations encountered during the research. The recommendations are directed toward four major stakeholder groups—learners, instructors and curriculum designers, institutional administrators and placement teams, and EdTech developers—because the effectiveness of the proposed AI-based skill gap analysis system depends on coordinated action across these levels. The limitations section acknowledges methodological, technical, ethical, and practical constraints that influence the interpretation and generalization of the results. The chapter also outlines prioritized directions for future research and large-scale deployment.

7.2 Recommendations

The recommendations are organized by stakeholder group and reflect both short-term actionable steps and longer-term strategic improvements.

7.2.1 For Learners (Students and Early-Career Professionals)

1. Adopt diagnostic-first study habits.

Learners should begin preparation for any new topic or job role with short diagnostic assessments instead of relying on self-perception. Diagnostics accurately identify weak concepts and ensure that study time is invested where it produces the greatest improvement.

2. Follow prerequisite-aware micro-learning paths.

Rather than enrolling in long, generic courses, learners should follow short, sequenced micro-modules that address prerequisite gaps before advancing to higher-level topics.

3. Practice active retrieval and spaced repetition.

Learning should be reinforced through frequent low-stakes quizzes and spaced revision, which the study showed to be strongly associated with higher mastery gains.

4. Monitor response time and confidence.

Learners should use behavioral indicators such as long response times or low confidence to identify concepts that require deeper revision.

7.2.2 For Instructors and Curriculum Designers

5. Use cohort analytics to guide intervention.

Teachers should regularly review concept-level dashboards to identify topics that most students struggle with and plan targeted remedial sessions.

6. Integrate diagnostic MCQs into formative assessment.

Concept-tagged MCQs should be embedded into coursework so that learning data continuously feeds both instruction and analytics.

7. Apply mastery thresholds for progression.

Students should demonstrate sufficient mastery of prerequisite concepts before being allowed to progress to advanced modules or projects.

8. Develop explicit prerequisite maps.

Departments should document and publish prerequisite relationships between concepts and modules to support both teaching and automated recommendation systems.

7.2.3 For Institutional Administrators and Placement Teams

9. Embed diagnostics into placement preparation.

Role-readiness scores should be used alongside interviews and portfolios to guide targeted upskilling before placement.

10. Fund short diagnostic-driven bridging programs.

Institutions should replace generic bootcamps with short, data-driven skill-bridging programs focused on the most common gaps.

11. Train faculty in data-informed pedagogy.

Faculty development programs should help instructors interpret dashboards and translate insights into teaching actions.

12. Ensure ethical data governance.

Clear policies for data privacy, consent, and access must be established to maintain trust and compliance.

7.2.4 For EdTech Developers and Researchers

13. Prioritize explainable recommendations.

Users should be shown why a resource was recommended to improve trust and engagement.

14. Emphasize free and open resources.

Recommendation systems should prioritize high-quality open educational resources to ensure equitable access.

15. Improve assessment quality and tagging.

High-quality items and accurate concept tagging are essential for reliable diagnostics.

16. Use controlled experimentation.

New features should be validated using A/B testing and controlled trials to ensure scientific rigor.

7.2.5 Cross-Stakeholder and Policy Recommendations

17. Standardize concept taxonomies across institutions.

Shared concept frameworks enable cross-course and cross-institution analytics.

18. Measure learning by mastery rather than completion.

Competency-based metrics better reflect real skill development.

19. Calibrate role thresholds with industry.

Employers should help define realistic benchmarks for role readiness.

20. Plan for long-term system maintenance.

Regular updating of items, models, and resources is necessary to prevent performance drift.

7.3 Limitations of the Study

7.3.1 Methodological Limitations

1. MCQ-based assessments cannot fully capture complex problem-solving, programming fluency, or creativity.
2. The sample was institution-specific, limiting external validity.
3. Engagement measures were partly self-reported, introducing possible bias.

7.3.2 Technical and Modelling Limitations

4. Some advanced concepts had limited real data.
5. Models may reflect bias present in training data and may drift over time.
6. Interpretable models may sacrifice slight predictive accuracy compared to black-box approaches.

7.3.3 Practical and Operational Limitations

7. The system was tested at proof-of-concept scale rather than full institutional scale.

8. Maintaining up-to-date learning resources requires continuous effort.

7.3.4 Ethical and Privacy Limitations

9. Privacy constraints limit deep personalization.

10. There is a risk of over-reliance on automated scores in high-stakes decisions.

7.4 Prioritized Future Work

Future research should focus on integrating coding-based assessments, improving engagement tracking, expanding concept taxonomies, implementing adaptive testing, conducting long-term outcome studies, strengthening fairness controls, and validating role-readiness scores with employers.

7.5 Roadmap for Institutional Adoption

A phased approach—pilot, evaluate, and scale—should be followed, supported by governance structures, dedicated resourcing, stakeholder training, and integration with existing LMS and placement systems.

7.6 Final Remarks

The proposed system offers a practical, evidence-based approach to diagnosing and remediating technical skill gaps. While it cannot replace high-quality teaching, projects, and mentorship, it provides a powerful data-driven foundation to support them.

7.7 Chapter Summary

This chapter presented actionable recommendations for key stakeholders, transparently acknowledged limitations, and outlined a roadmap for future development and institutional adoption.

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APPENDIX

APPENDIX A – Sample MCQ Structure

This appendix presents an illustrative sample of the Multiple Choice Question (MCQ) format used in the diagnostic assessments. Each question is associated with a concept and difficulty level to support fine-grained mastery estimation.

Question ID	Concept	Difficulty	Question	Correct Answer
Q101	Probability	Medium	What is $P(A \cap B)$ when A and B are independent?	$P(A) \times P(B)$
Q205	Linear Regression	Hard	What does the regression coefficient represent?	Slope

APPENDIX B – Concept Taxonomy (Excerpt)

This table shows a portion of the knowledge graph used by the system, indicating prerequisite relationships among concepts.

Domain	Concept	Prerequisite(s)
Statistics	Mean	None
Statistics	Variance	Mean
Machine Learning	Linear Regression	Mean, Variance
Machine Learning	Classification	Probability, Regression

APPENDIX C – Role–Skill Mapping

This mapping defines the core competencies required for each target job role and is used to compute role-readiness scores.

Job Role	Required Concepts
Data Analyst	SQL, Statistics, Data Cleaning
ML Engineer	Linear Algebra, Probability, Optimization

APPENDIX D – System Workflow

The operational workflow of the AI-based Skill Gap Analysis and Learning Recommendation System is summarized below:

1. The learner completes an MCQ-based diagnostic assessment.
2. The AI model estimates concept-level mastery.
3. A Skill Gap Index is computed for each concept.
4. The knowledge graph identifies missing prerequisite concepts.
5. A personalized learning pathway is generated.
6. The learner studies the recommended learning resources.
7. A post-test measures learning improvement.

APPENDIX E – Sample Learner Diagnostic Report

An example of the learner-level output generated by the system is shown below.

Concept	Mastery Probability	Skill Gap Level
Probability	0.42	High
Regression	0.78	Low
SQL	0.85	None

